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(54) **METHOD FOR PROVIDING INFORMATION FOR PREDICTING RISK GROUP FOR DEVELOPING ALZHEIMER'S DISEASE OR RISK GROUP FOR EARLY ONSET OF ALZHEIMER'S SYMPTOMS, OR RISK GROUP FOR DEVELOPING AMNESTIC MILD COGNITIVE IMPAIRMENT AND/OR PET-POSITIVE RISK GROUP FOR AMYLOID β DEPOSITION, BASED ON EUROPEAN POPULATION DATA**

(71) Applicants: **SAMSUNG LIFE PUBLIC WELFARE FOUNDATION**, Seoul (KR); **RESEARCH & BUSINESS FOUNDATION SUNGKYUNKWAN UNIVERSITY**, Gyeonggi-do (KR); **DONGGUK UNIVERSITY INDUSTRY-ACADEMIC COOPERATION**, Seoul (KR)

(72) Inventors: **Hee Jin KIM**, Seoul (KR); **Hong Hee WON**, Seoul (KR); **Hang Rai KIM**, Gyeonggi-do (KR); **Sang Hyuk JUNG**, Seoul (KR)

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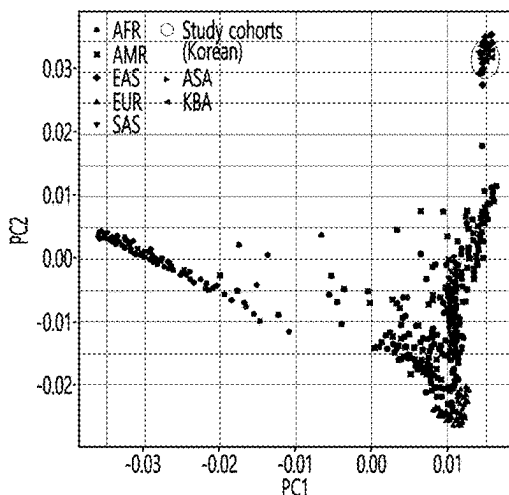
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(57)

ABSTRACT

Embodiments of the present disclosure herein relate to a method for providing information for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, or a risk group for developing amnesic mild cognitive impairment and/or a positron emission tomography (PET)-positive risk group for amyloid β deposition, based on European population data. In an embodiment, the method makes it possible to accurately predict a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, or a risk group for developing amnesic mild cognitive impairment and/or a positron emission tomography (PET)-positive risk group for amyloid β deposition by using only at least 11 single-nucleotide polymorphisms, and the ability to predict the risk groups is further enhanced when up to and at most 39 additional single-nucleotide polymorphisms are used.

A.



B.

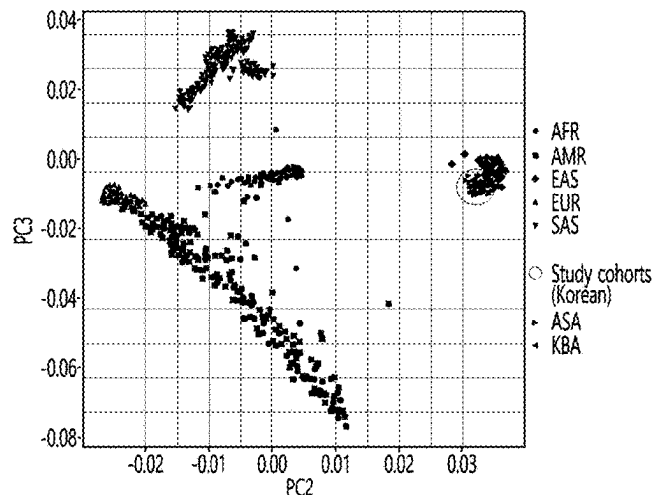


Fig. 1A

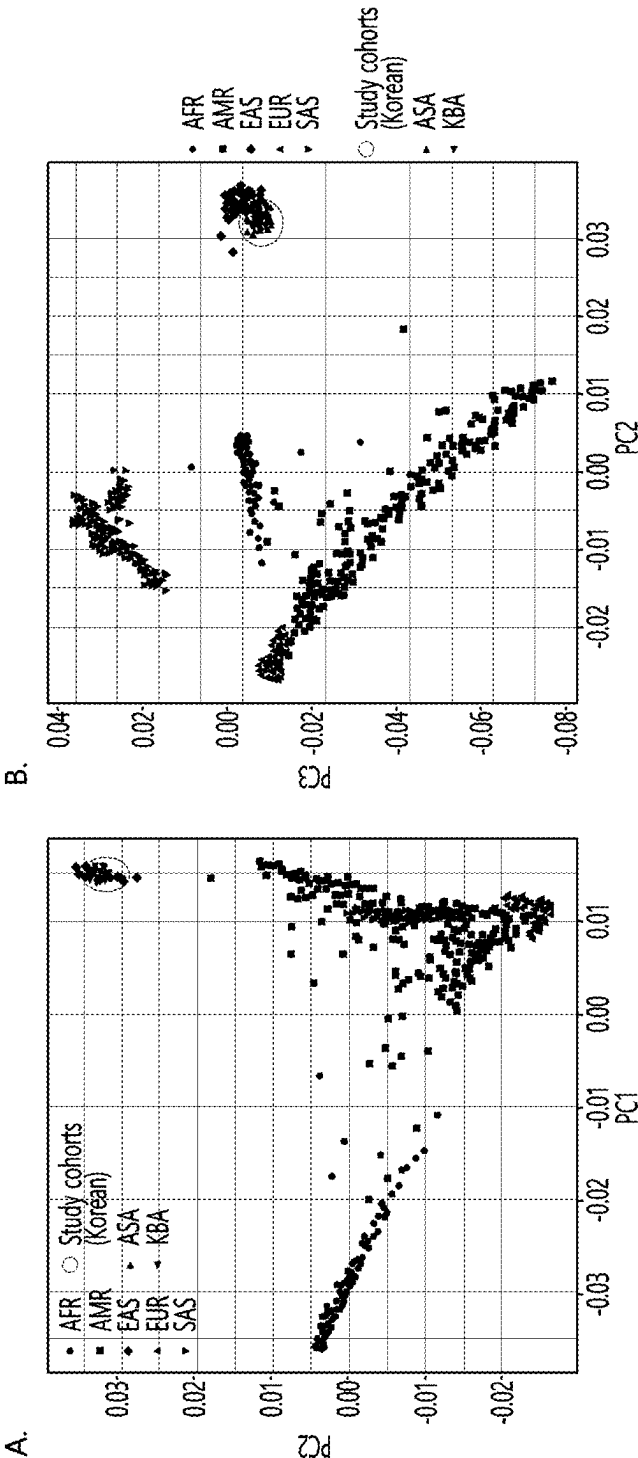


Fig. 1B

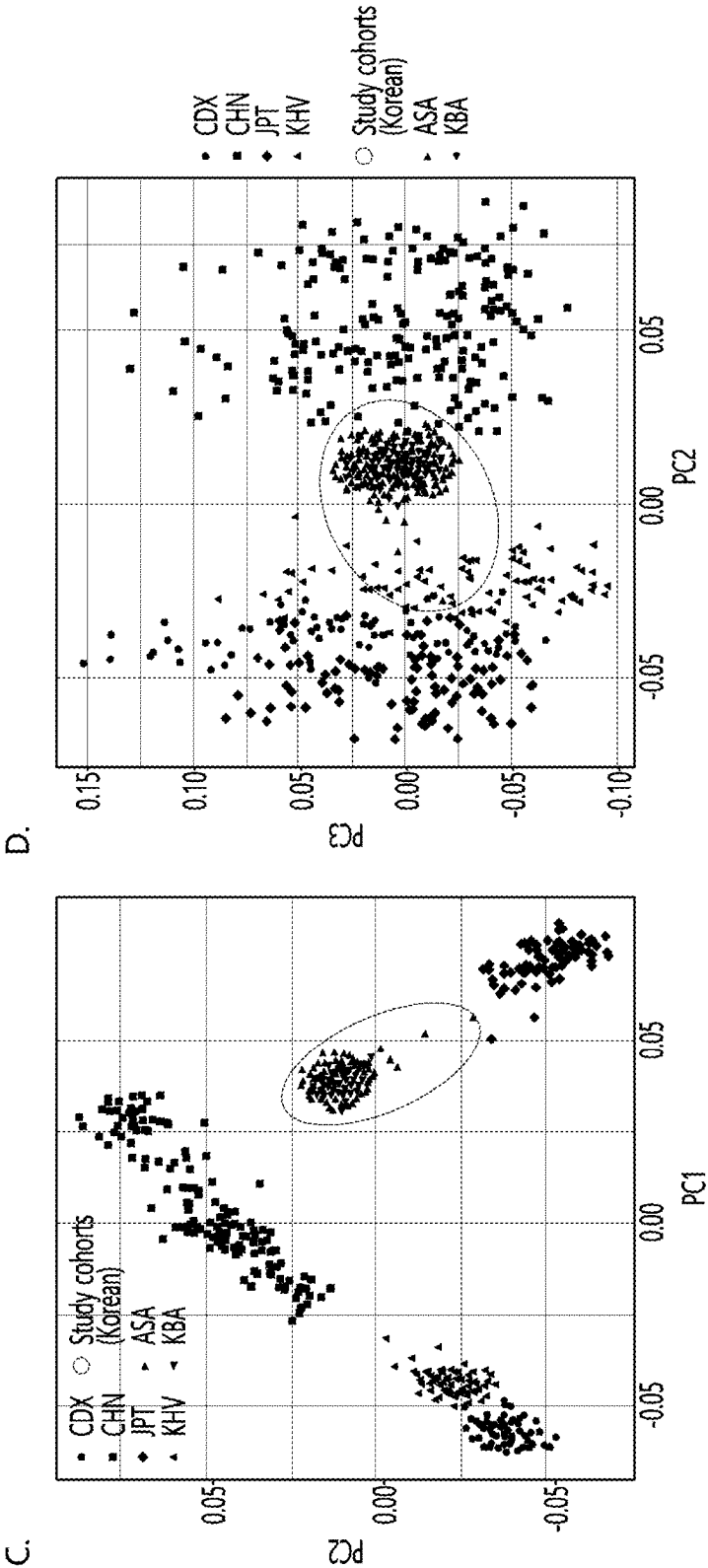


Fig. 2

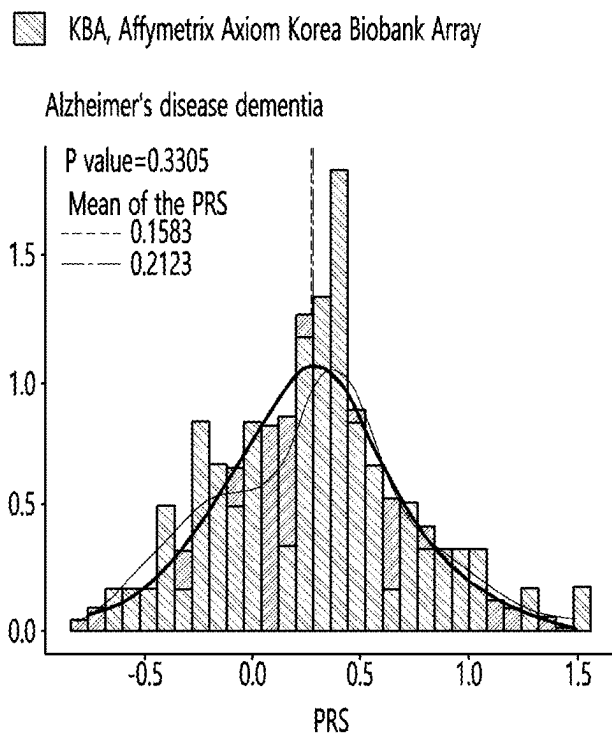
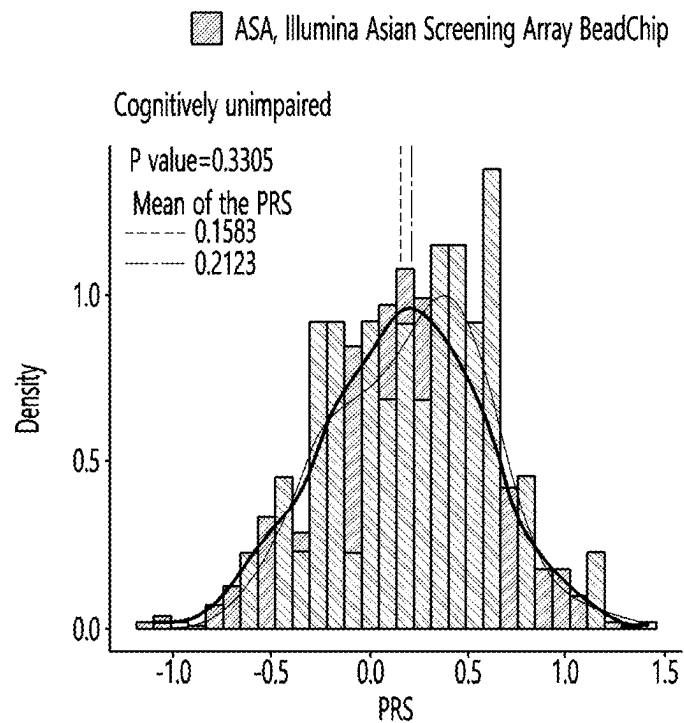


Fig. 3

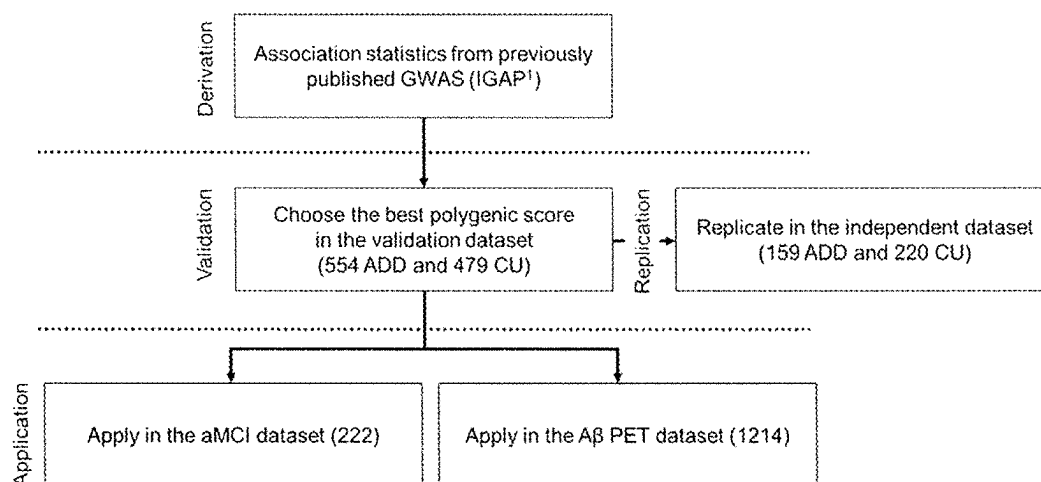


Fig. 4

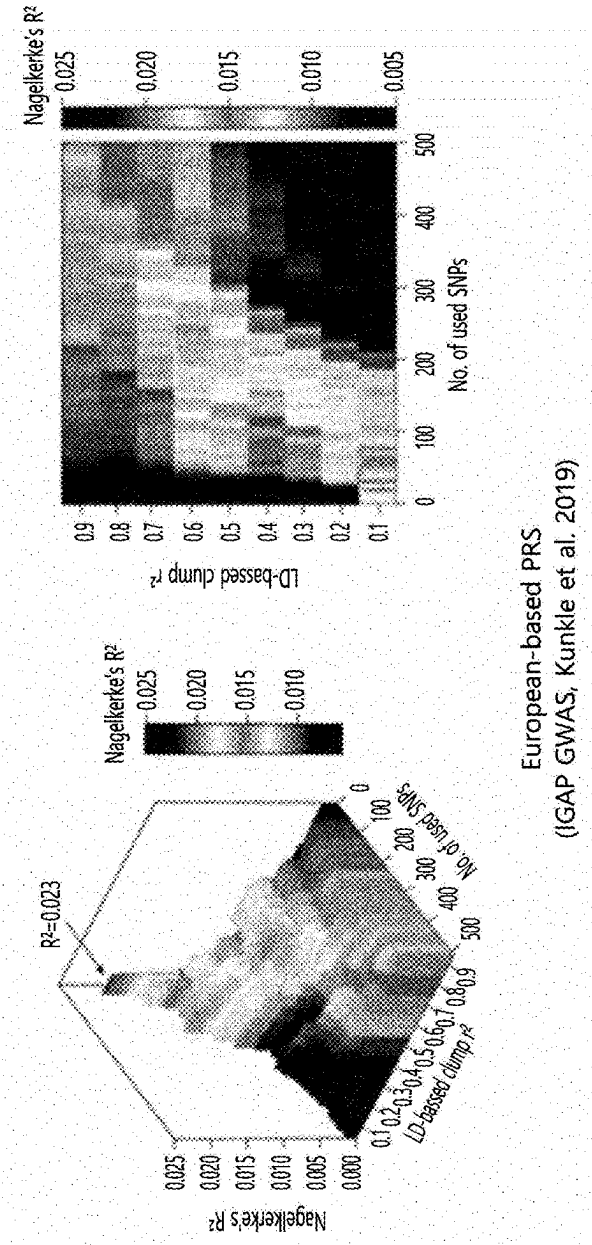


Fig. 5

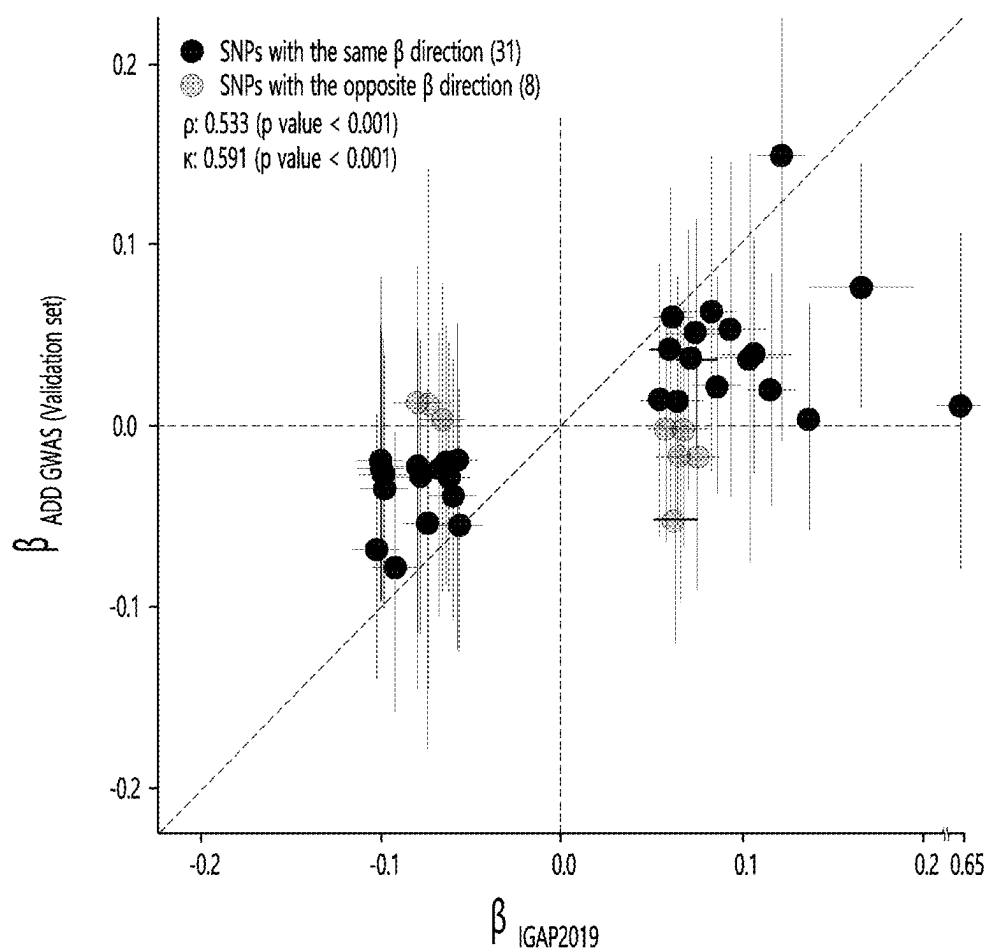


Fig. 6A

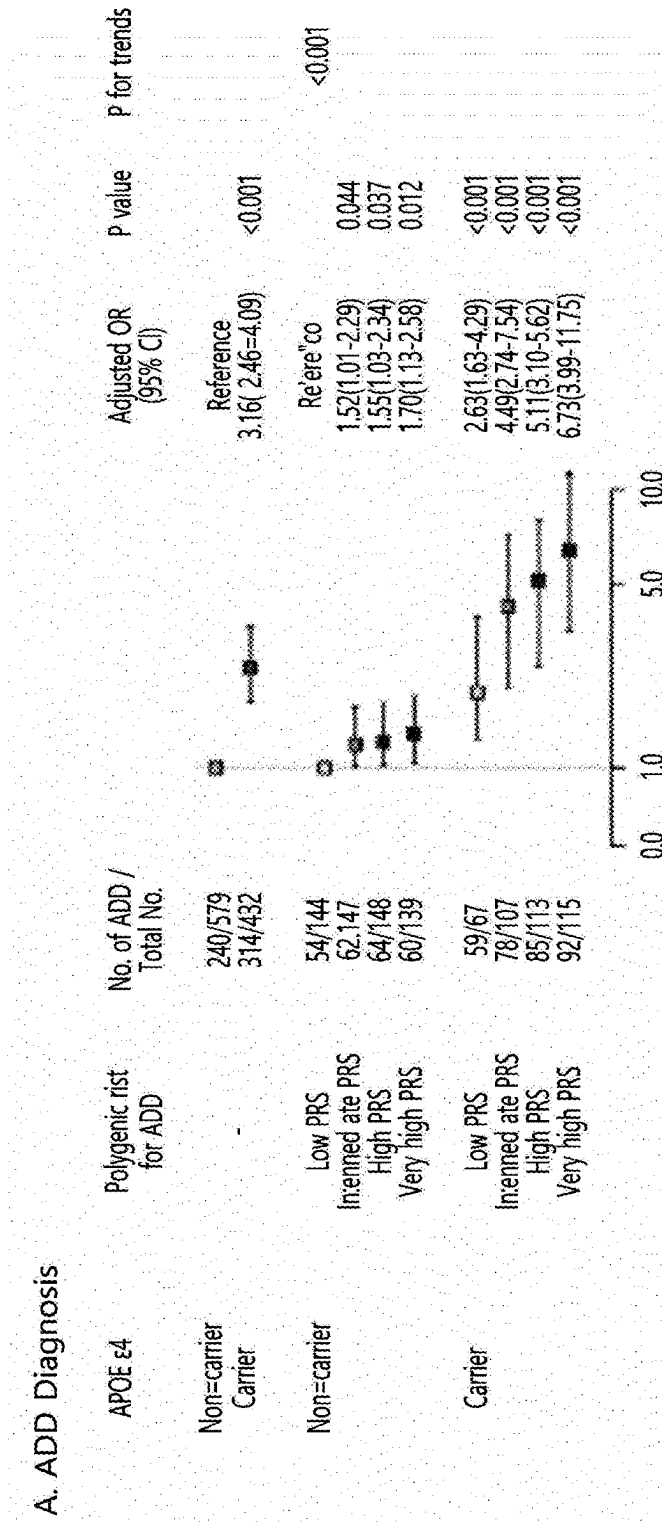


Fig. 6B

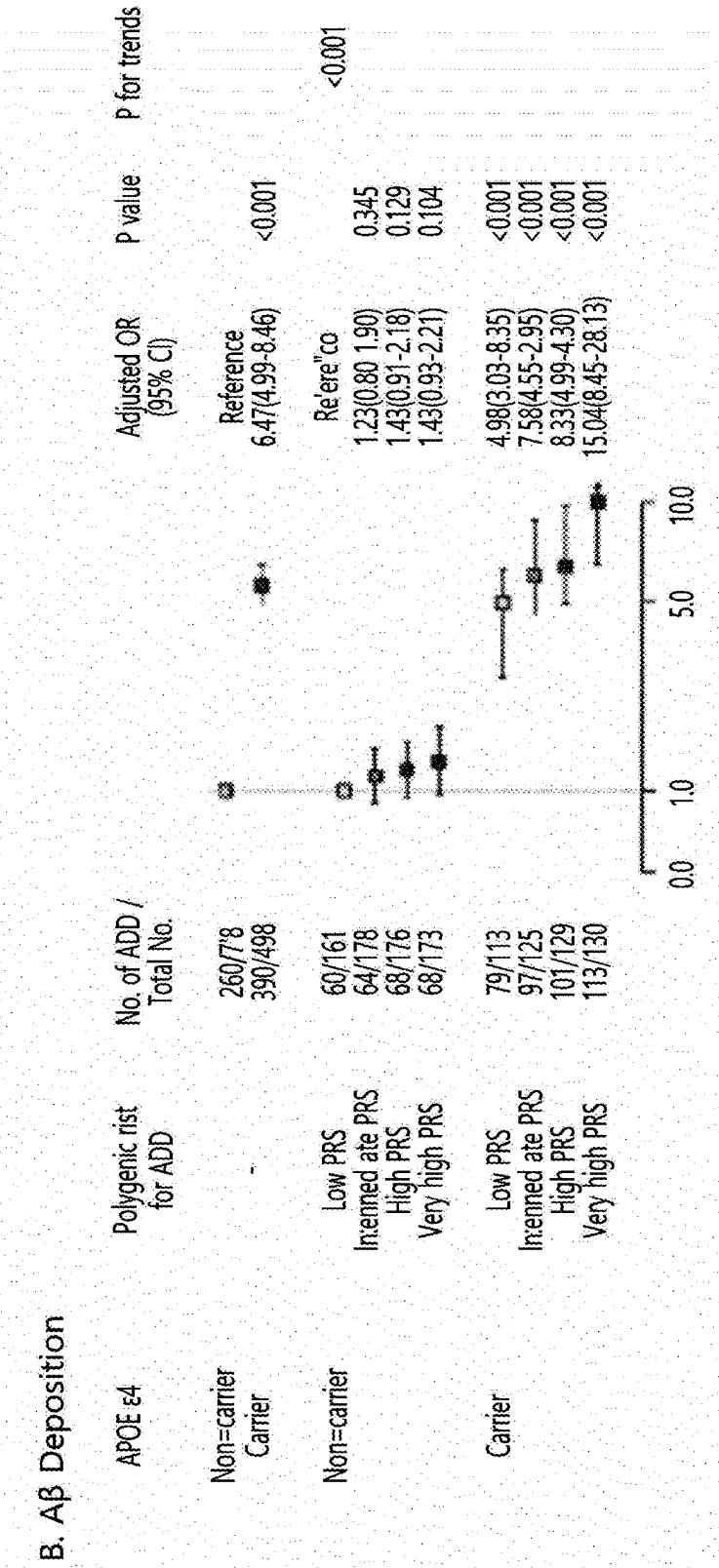
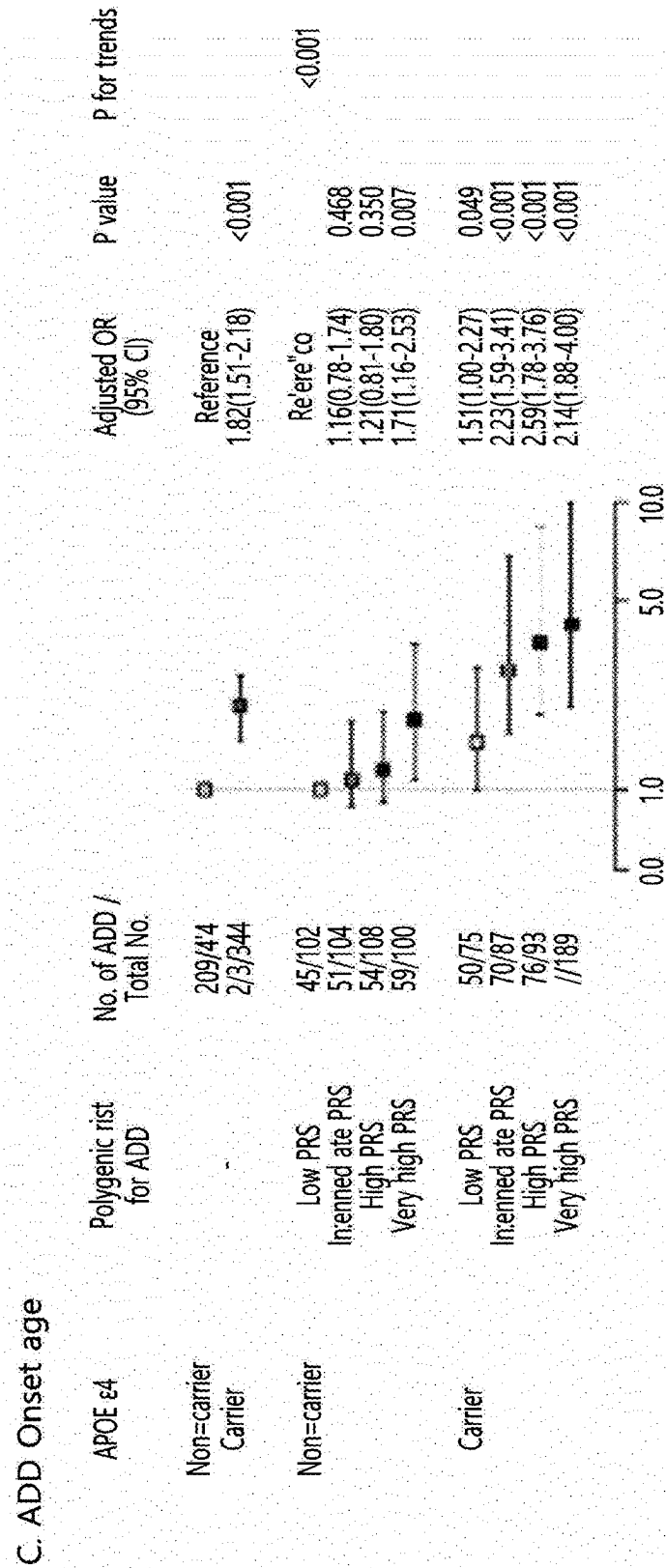


Fig. 6C



**METHOD FOR PROVIDING INFORMATION
FOR PREDICTING RISK GROUP FOR
DEVELOPING ALZHEIMER'S DISEASE OR
RISK GROUP FOR EARLY ONSET OF
ALZHEIMER'S SYMPTOMS, OR RISK
GROUP FOR DEVELOPING AMNESTIC
MILD COGNITIVE IMPAIRMENT AND/OR
PET-POSITIVE RISK GROUP FOR AMYLOID
B DEPOSITION, BASED ON EUROPEAN
POPULATION DATA**

TECHNICAL FIELD

[0001] Embodiments of the present disclosure herein relate to a method for providing information for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, or a risk group for developing amnestic mild cognitive impairment and/or a PET-positive risk group for amyloid β deposition, based on European population data.

BACKGROUND ART

[0002] Alzheimer's disease is the leading cause of dementia, affecting about 50 million people worldwide, and the number of patients with the disease is expected to triple by 2050 due to population aging. In particular, the disease is problematic in East Asia, where the population is aging rapidly. Almost a quarter of people with dementia are estimated to live in East Asia, and that number is expected to double every 20 years.

[0003] The pathological process of Alzheimer's disease begins long before clinical dementia develops. Therefore, it is very important for potential prevention and treatment strategies to identify individuals at high risk of developing Alzheimer's disease. Because the heritability of Alzheimer's disease is estimated to be 60 to 80%, genetic information may be used to identify individuals at high risk of developing Alzheimer's disease. Previous studies have demonstrated that polygenic risk scores (PRSs), which summarize the genetic effects of single nucleotide polymorphisms (SNPs) identified in genome-wide association studies (GWASs), can help distinguish individuals at high genetic risk for Alzheimer's disease.

[0004] However, previous genetic studies have been conducted primarily on European populations. Therefore, the generalizability of the PRS for non-European populations is not yet known. Recent studies have investigated the ability to predict European ancestry-derived PRS in non-European ancestry samples for various phenotypes. The PRS for Alzheimer's disease derived from European populations was evaluated in black and Caribbean Hispanic populations, but not in Hispanics. However, the performance of the PRS for Alzheimer's disease has not yet been evaluated in Asian populations.

[0005] Therefore, the present inventors have conducted studies in order to validate the possibility of inter-ethnic transfer of PRS for Alzheimer's disease in a Korean population using summarized statistics from a previous large-scale GWAS on a European population. The present inventors reproduced the next study results in an independent cohort of the Korean population, and assessed whether polygenic risk scores could be applied to predict Alzheimer's disease dementia (ADD), amnestic mild cognitive impairment (aMCI), or amyloid β (A β) deposition.

DISCLOSURE

Technical Problem

[0006] An object of the present disclosure is to develop and provide a method for providing information for predicting a risk group for developing Alzheimer's disease or a risk group for early onset of Alzheimer's symptoms, or a risk group for developing amnestic mild cognitive impairment and a positron emission tomography (PET)-positive risk group for amyloid β deposition using a polygenic risk score (PRS) for Korean people, based on a method for predicting a risk group for Alzheimer's disease dementia using an existing polygenic risk score (PRS) for European people.

Technical Solution

[0007] An embodiment of the present disclosure provides a method for providing information for predicting a risk group for developing Alzheimer's disease dementia (ADD) or a risk group for early onset of Alzheimer's symptoms, the method including: bringing a sample isolated from an individual in contact with a preparation capable of identifying the presence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs); and

[0008] determining the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms in the sample,

[0009] wherein the plurality of single-nucleotide polymorphisms includes rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0010] The method for providing information for predicting a risk group for developing Alzheimer's disease dementia (ADD) or a risk group for early onset of Alzheimer's symptoms was derived by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing a predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0011] In an embodiment, the method may predict a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0012] The term "individual" may refer to a mammal, for example, a mouse, rat, cat, guinea pig, hamster, dog, monkey, chimpanzee, human, and the like, and specifically a human.

[0013] The sample may be blood, plasma, serum, tissue, cells, lymphatic fluid, bone marrow fluid, saliva, ocular fluid, semen, brain extract, spinal fluid, synovial fluid, thymic fluid, ascitic fluid, amniotic fluid, cell tissue fluid, and cell culture fluid, and may be specifically blood, plasma, serum, tissue, cells, lymphatic fluid, bone marrow fluid, cell tissue fluid, and cell culture fluid, and more specifically blood or saliva.

[0014] The preparation may be selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence including the single-nucleotide polymorphism or a protein encoded by the base sequence.

[0015] The term “primer” refers to a nucleic acid sequence that can form complementary base pairs with a template strand and serve as a starting point for template strand copying, and may be, for example, a nucleic acid sequence of 5 to 50 amino acids. The primer is usually synthesized, but may also be used on naturally occurring nucleic acids. The sequence of the primer does not necessarily have to be exactly the same as the sequence of the template, but it must be sufficiently complementary to hybridize with the template.

[0016] The term “probe” refers to a material capable of specifically binding to a target material to be detected in a sample, and refers to a material in a sample of which the presence of a target material can be specifically confirmed through the binding.

[0017] The term “aptamer” refers to a small single-stranded nucleic acid (DNA or RNA) fragment that has the characteristics of being able to bind with high affinity and specificity to various types of substances, from low molecular weight compounds to proteins, and may be, for example, a single-stranded nucleic acid fragment consisting of 10 to 60 nucleotides.

[0018] The term “antibody” refers to a substance that specifically binds to an antigen to cause an antigen-antibody reaction, and may be a chimeric antibody, a humanized antibody, a human antibody, a synthetic antibody and/or an affinity matured antibody.

[0019] The term “peptide” refers to a polymer consisting of two or more amino acids linked together through amide bonds (or peptide bonds).

[0020] The term “single-nucleotide polymorphisms (SNPs)” refers to the presence of two or more alleles at a single gene locus, where only a single nucleotide is different at a polymorphic site.

[0021] The term “Alzheimer’s disease dementia” refers to the dementia symptoms induced by Alzheimer’s disease. Alzheimer’s disease described above is the most common degenerative brain disease that causes dementia, which was first reported by Dr. Alzheimer in Germany, and it is known that as Alzheimer’s disease progresses, overall cognitive functions including memory gradually weaken.

[0022] The term “onset” refers to the beginning of a disease, and the term “early onset of symptoms” refers to the manifestation of various states or forms appearing when one is suffering from a disease at the early stage of the onset of the disease.

[0023] The determining of the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms in the sample may include determining the presence or absence of risk alleles of the plurality of single-nucleotide polymorphisms by detecting a nucleic acid (for example, DNA, RNA, and the like) and/or a protein in the sample to analyze a genotype. Specifically, the genotype may be analyzed by detecting DNA in a sample using, for example, an Illumina Asian Screening Array Bead Chip (ASA chip, CA).

[0024] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444,

rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0025] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the method has better performance when predicting a risk group for developing Alzheimer’s disease dementia or a risk group for early onset of Alzheimer’s symptoms, and the higher the number of single-nucleotide polymorphisms that are further included, the better the predictive performance of the method may be.

[0026] In an embodiment, the method may further include obtaining a score for a single-nucleotide polymorphism by assigning a score of 1 to a single-nucleotide polymorphism determined to indicate the presence of a risk allele in the sample among the plurality of single-nucleotide polymorphisms, wherein among the plurality of single-nucleotide polymorphisms, a single-nucleotide polymorphism determined to be absent from the sample is assigned a score of 0.

[0027] In an embodiment, the method may further include obtaining a first polygenic risk score (PRS) value by multiplying the assigned score for the single-nucleotide polymorphism by a coefficient (β) assigned for each of the following single-nucleotide polymorphisms, and adding all the multiplied values, and

[0028] the coefficient of rs6733839 may be 0.1693, the coefficient of rs1582763 may be -0.1232 , the coefficient of rs679515 may be 0.1508, the coefficient of rs1532276 may be -0.1266 , the coefficient of rs3851179 may be -0.1198 , the coefficient of rs1752684 may be 0.1432, the coefficient of rs56201148 may be -0.1137 , the coefficient of rs67472071 may be -0.0981 , the coefficient of rs35832505 may be 0.1213, the coefficient of rs12151021 may be 0.1071, the coefficient of rs73223431 may be 0.0936, the coefficient of rs12358692 may be 0.6429, the coefficient of rs11218343 may be -0.2053 , the coefficient of rs6014724 may be -0.1319 , the coefficient of rs17125924 may be 0.1222, the coefficient of rs35695568 may be 0.1152, the coefficient of rs11767557 may be -0.1028 , the coefficient of rs3795065 may be 0.0968, the coefficient of rs3752786 may be 0.0964, the coefficient of rs11623019 may be 0.0913, the coefficient of rs12590654 may be -0.0906 , the coefficient of rs11039165 may be 0.0894, the coefficient of rs28482811 may be -0.0872 , the coefficient of rs3135348 may be 0.0837, the coefficient of rs9381563 may be 0.0821, the coefficient of rs9268112 may be 0.0815, the coefficient of rs3865444 may be -0.0804 , the coefficient of rs11230227 may be 0.0792, the coefficient of rs9271375 may be -0.0789 , the coefficient of rs2293579 may be 0.0771, the coefficient of rs7831810 may be -0.0765 , the coefficient of rs11168036 may be 0.0754, the coefficient of rs598561 may be 0.0747, the coefficient of rs3017432 may be -0.0735 , the coefficient of rs2526378 may be 0.0717, the coefficient of rs8111708 may be 0.0696, the coefficient of rs7805776 may be -0.0695 , and the coefficient of rs12197146 may be -0.0674 .

[0029] The term “polygenic risk score (PRS)” refers to a score that predicts the risk of developing a disease by evaluating various genetic factors associated with one disease.

[0030] The term “gene” refers to a segment of DNA involved in producing a polypeptide chain. The DNA segment may include intervening sequences (introns) between individual coding segments (exons), as well as the regions preceding and following the coding region (leader or trailer) involved in the transcription/translation of a gene product and the regulation of transcription/translation.

[0031] In an embodiment, the method may further include determining that when the first PRS value is higher than the first PRS value of an individual not having Alzheimer’s disease dementia, the individual is in a high risk group for developing Alzheimer’s disease dementia or in a high risk group for early onset of Alzheimer’s symptoms.

[0032] In an embodiment, when the plurality of single-nucleotide polymorphisms further include rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146, the obtaining of the first PRS value may include calculating, for example, the first PRS value using the following Mathematical Formula 1:

[Mathematical Formula 1]

the first PRS value (39 SNPs) =

$$\begin{aligned} &rs12358692 \times 0.6429 + rs11218343 \times -0.2053 + rs6733839 \times 0.1693 + \\ &rs679515 \times 0.1508 + rs1752684 \times 0.1432 + rs6014724 \times -0.1319 + \\ &rs6805148 \times -0.1293 + rs1532276 \times -0.1266 + rs1582763 \times -0.1232 + \\ &rs17125924 \times 0.1222 + rs35832505 \times 0.1213 + rs3851179 \times -0.1198 + \\ &rs35695568 \times 0.1152 + rs56201148 \times -0.1137 + rs12151021 \times 0.1071 + \\ &rs11767557 \times -0.1028 + rs67472071 \times -0.0981 + rs3795065 \times 0.0968 + \\ &rs3752786 \times 0.0964 + rs73223431 \times 0.0936 + rs11623019 \times 0.0913 + \\ &rs12590654 \times -0.0906 + rs11039165 \times 0.0894 + rs28482811 \times -0.0872 + \\ &rs3135348 \times 0.0837 + rs9381563 \times 0.0821 + rs9268112 \times 0.0815 + \\ &rs3865444 \times -0.0804 + rs11230227 \times 0.0792 + rs9271375 \times -0.0789 + \\ &rs2293579 \times 0.0771 + rs7831810 \times -0.0765 + rs11168036 \times 0.0754 + \\ &rs598561 \times 0.0747 + rs3017432 \times -0.0735 + rs2526378 \times 0.0717 + \\ &rs8111708 \times 0.0696 + rs7805776 \times -0.0695 + rs12197146 \times -0.0674. \end{aligned}$$

[0033] In an embodiment, the method may further include identifying one or more indicators selected from the group consisting of the individual’s age, sex, years of education, and APOE genotype.

[0034] In an embodiment, when the method further includes identifying one or more indicators selected from the group consisting of the individual’s age, sex, years of education, and APOE genotype, the method has better performance when predicting a risk group for developing Alzheimer’s disease dementia or a risk group for early onset of Alzheimer’s symptoms, and the higher the number of indicators identified, the better the predictive performance of the method may be.

[0035] The term “Apolipoprotein (APOE) genotype” refers to the genotype of a gene encoding Apolipoprotein E. APOE is a gene encoding the constituent proteins of lipoproteins, which are normal components of plasma such as high density lipoprotein (HDL), low density lipoprotein (LDL), and very low density lipoprotein (VLDL), and is located on chromosome 19 and plays a key role in regulating fat metabolism after damage to the central and peripheral nervous systems as a fat transporter. The APOE gene has three allelic genotypes: $\epsilon 2$, $\epsilon 3$, and $\epsilon 4$, and is classified into a total of six genotypes: $\epsilon 2/\epsilon 2$, $\epsilon 3/\epsilon 3$, $\epsilon 4/\epsilon 4$, $\epsilon 2/\epsilon 3$, $\epsilon 2/\epsilon 4$, and $\epsilon 3/\epsilon 4$ because every person inherits one allele of the APOE gene from each parent. Among the allelic genotypes, $\epsilon 2$ and $\epsilon 4$ are known to be associated with Alzheimer’s disease dementia, with the $\epsilon 2$ type known to decrease the risk of Alzheimer’s disease dementia and the $\epsilon 4$ type known to increase the risk of Alzheimer’s disease dementia.

[0036] In an embodiment, the method may further include obtaining a score for each indicator by assigning a score (natural number) based on the number of years of education in the case of the age and years of education among the indicators of the individual,

[0037] assigning a score of 1 for males and a score of 2 for females in the case of sex among the indicators of the individual, and

[0038] assigning a score of 0 for $\epsilon 2/\epsilon 2$, $\epsilon 2/\epsilon 3$, and $\epsilon 3/\epsilon 3$ and a score of 1 for $\epsilon 2/\epsilon 4$, $\epsilon 3/\epsilon 4$, and $\epsilon 4/\epsilon 4$ in the case of APOE genotype among the indicators of the individual.

[0039] When the indicator is age in the obtaining of the score for each of the indicators, a score (natural number) is assigned based on the number of years, and for example, when the age is 72 years and 6 months, a score of 72 may be assigned.

[0040] When the indicator is years of education in the obtaining of the score for each of the indicators, a score (natural number) is assigned based on the number of years, and for example, when the years of education are 11 years and 2 months, a score of 11 may be assigned.

[0041] When the indicator is sex in the obtaining of the score for each of the indicators, if gender, a score of 1 and a score of 2 may be assigned for males and females, respectively.

[0042] The method may further include obtaining a score for each indicator by assigning a score of 0 for $\epsilon 2/\epsilon 2$, $\epsilon 2/\epsilon 3$, and $\epsilon 3/\epsilon 3$ and a score of 1 for $\epsilon 2/\epsilon 4$, $\epsilon 3/\epsilon 4$, and $\epsilon 4/\epsilon 4$ in the case of APOE genotype among the indicators of the individual.

[0043] In an embodiment, the method may further include obtaining a second PRS value by multiplying the assigned score for each indicator by a coefficient (β) assigned for each of the following indicators, and adding the first PRS value and a coefficient (β) assigned for the following first PRS value to the multiplied values, and

[0044] the coefficient of the age may be 0.02879, the coefficient of the sex may be 0.03618, the coefficient of the years of education may be -0.02615 , the coefficient of the APOE genotype may be 1.3712, and the coefficient of the first PRS value may be 0.66119.

[0045] The obtaining of the second PRS value may calculate, for example, the second PRS value using a mathematical formula represented by the following Mathematical Formula 2:

[Mathematical Formula 2]

Second PRS value (including 4 factors) =

$$\text{first PRS}^*0.66119 + \text{age}^*0.02879 + \text{sex}^*0.03618 + \\ \text{years of education}^* - 0.02615 + \text{APOE}\epsilon 4^*1.37120.$$

[0046] In an embodiment, the method may further include determining that when the second PRS value is higher than the second PRS value of an individual not having Alzheimer's disease dementia, the individual is in a high risk group for developing Alzheimer's disease dementia or in a high risk group for early onset of Alzheimer's symptoms.

[0047] Another embodiment of the present disclosure provides a method for providing information for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, the method including: bringing a sample isolated from an individual in contact with a preparation capable of identifying the presence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs); and

[0048] determining the presence or absence of risk alleles of the plurality of single-nucleotide polymorphisms,

[0049] wherein the plurality of single-nucleotide polymorphisms include rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3752786, rs11623019, rs12590654, rs28482811, rs9381563, rs11230227, rs9271375, rs7831810, rs11168036, rs598561, rs3017432, rs8111708, rs7805776, and rs12197146.

[0050] The "individual," "sample," "single-nucleotide polymorphism," "preparation," "Alzheimer's disease dementia," and the like may be within the above-described scope.

[0051] The method for providing information for predicting a risk group for developing Alzheimer's disease dementia (ADD) or a risk group for early onset of Alzheimer's symptoms was derived by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0052] In an embodiment, the method may predict a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms with excellent accuracy by confirming the presence or absence of risk alleles of 31 single-nucleotide polymorphisms in a sample.

[0053] Still another embodiment of the present disclosure provides a composition for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, including a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual,

[0054] wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0055] The "individual," "sample," "single-nucleotide polymorphism," "Alzheimer's disease dementia," and the like may be within the above-described scope.

[0056] The composition for predicting a risk group for developing Alzheimer's disease dementia (ADD) or a risk group for early onset of Alzheimer's symptoms was invented by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing a predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0057] In an embodiment, the composition may predict a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0058] The preparation may be selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence including the single-nucleotide polymorphism or a protein encoded by the base sequence.

[0059] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0060] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the composition has better performance when predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, and the higher the number of single-nucleotide polymorphisms is, the better the predictive performance of the composition may be.

[0061] In an embodiment, the composition may further include a preparation capable of identifying the APOE genotype of the individual in the sample.

[0062] In an embodiment, when the composition further includes a preparation capable of identifying the APOE genotype of the individual in the sample, the composition has better performance when predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms.

[0063] Yet another embodiment of the present disclosure provides a kit for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms, including the composition.

[0064] The "Alzheimer's disease dementia" may be within the above-described scope.

[0065] The kit may be a reverse transcription polymerase chain reaction (RT-PCR) kit or a DNA chip kit.

[0066] The kit for predicting a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms may additionally include one or more other constituent compositions, solutions or devices suitable for an analytical method.

[0067] As an example, the kit may be a diagnostic kit characterized by including essential elements required for carrying out a reverse transcription polymerase reaction, and in addition to a primer capable of specifically binding to the APOE gene or a base sequence including the single-nucleotide polymorphisms, the kit may include a test tube or another suitable container, a reaction buffer (pH and magnesium concentration are various), deoxynucleotides (dNTPs), an enzyme such as Taq-polymerase and a reverse transcriptase, DNase, RNase inhibitor DEPC-water, sterile water, and the like.

[0068] As another example, the kit may be a diagnostic kit characterized by including essential elements required for carrying out a DNA chip. The DNA chip kit may include a substrate to which cDNA or oligonucleotides corresponding to genes or fragments thereof are attached, as well as a reagent, a preparation, an enzyme, and the like for producing a fluorescently labeled probe. Further, the substrate may include a cDNA or oligonucleotide corresponding to a control gene or a fragment thereof.

[0069] In an embodiment, the kit may predict a risk group for developing Alzheimer's disease dementia or a risk group for early onset of Alzheimer's symptoms with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0070] Yet another embodiment of the present disclosure provides a method for providing information for predicting a risk group for developing amnesic mild cognitive impairment, the method including: bringing a sample isolated from an individual in contact with a preparation capable of identifying the presence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs); and

[0071] determining the presence or absence of risk alleles of the plurality of single-nucleotide polymorphisms,

[0072] wherein the plurality of single-nucleotide polymorphisms include rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0073] The "individual," "sample," "single-nucleotide polymorphism," "preparation," and the like may be within the above-described scope.

[0074] The term "amnesic mild cognitive impairment" refers to a type of mild cognitive impairment (MCI) in which memory is primarily affected, and "mild cognitive impairment" refers to a type of mental deterioration with a clinical dementia rating (CDR) of less than 1.

[0075] The method for providing information for predicting a risk group for developing amnesic mild cognitive impairment was derived by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing a predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0076] In an embodiment, the method may predict a risk group for developing amnesic mild cognitive impairment

with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0077] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0078] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the method has better performance when predicting a risk group for developing amnesic mild cognitive impairment, and the higher the number of single-nucleotide polymorphisms that are further included, the better the predictive performance of the method may be.

[0079] In an embodiment, the method may further include identifying one or more indicators selected from the group consisting of the individual's age, sex, years of education, and APOE genotype.

[0080] In an embodiment, when the method further includes identifying one or more indicators selected from the group consisting of the individual's age, sex, years of education, and APOE genotype, the method has better performance when predicting a risk group for developing amnesic mild cognitive impairment, and the higher the number of indicators identified, the better the predictive performance of the method may be.

[0081] Yet another embodiment of the present disclosure provides a composition for predicting a risk group for developing amnesic mild cognitive impairment, including a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual,

[0082] wherein the plurality of SNPs includes rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0083] The "individual," "sample," "single-nucleotide polymorphism," "amnesic mild cognitive impairment," and the like may be within the above-described scope.

[0084] The composition for predicting a risk group for developing amnesic mild cognitive impairment was invented by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0085] In an embodiment, the composition may predict a risk group for developing amnesic mild cognitive impairment with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0086] The preparation may be selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence including the single-nucleotide polymorphism or a protein encoded by the base sequence.

[0087] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0088] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the composition has better performance when predicting a risk group for developing amnesic mild cognitive impairment, and the higher the number of single-nucleotide polymorphisms that are further included, the better the predictive performance of the composition may be.

[0089] In an embodiment, the composition may further include a preparation capable of identifying the APOE genotype of the individual in the sample.

[0090] In an embodiment, when the composition further includes a preparation capable of identifying the APOE genotype of the individual in the sample, the composition has better performance when predicting a risk group for developing amnesic mild cognitive impairment.

[0091] Yet another embodiment of the present disclosure provides a kit for predicting a risk group for developing amnesic mild cognitive impairment, including the composition.

[0092] The “amnesic mild cognitive impairment” and “kit” may be within the above-described scope.

[0093] In an embodiment, the kit may predict a risk group for developing amnesic mild cognitive impairment with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0094] Yet another embodiment of the present disclosure provides a method for providing information for predicting a positron emission tomography (PET)-positive risk group for amyloid β deposition, the method including: bringing a sample isolated from an individual in contact with a preparation capable of identifying the presence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs); and

[0095] determining the presence or absence of risk alleles of the plurality of single-nucleotide polymorphisms,

[0096] wherein the plurality of single-nucleotide polymorphisms include rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0097] The “individual,” “sample,” “single-nucleotide polymorphism,” “preparation,” and the like may be within the above-described scope.

[0098] The term “positron emission tomography (PET)-positive risk group for amyloid β deposition” refers to visual confirmation of the presence or absence of amyloid- β deposition through amyloid positron emission tomography (PET).

[0099] The term “positron emission tomography (PET)” refers to a technology in which a radiopharmaceutical which emits positrons is injected intravenously or inhaled into the body, and then the gamma rays generated by a positron

annihilation phenomenon are measured by a circular ring-shaped detector that surrounds the body as they pass through the body, and the distribution of positron-emitting nuclides within the body is processed by a computer to reconstruct an image.

[0100] The method for providing information for predicting a PET-positive risk group for amyloid β deposition was derived by using European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0101] In an embodiment, the method may predict a PET-positive risk group for amyloid β deposition with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0102] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0103] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the method has better performance when predicting a PET-positive risk group for amyloid β deposition, and the higher the number of single-nucleotide polymorphisms that are further included, the better the predictive performance of the method may be.

[0104] In an embodiment, the method may further include identifying one or more indicators selected from the group consisting of the individual’s age, sex, years of education, and APOE genotype.

[0105] In an embodiment, when the method further includes identifying one or more indicators selected from the group consisting of the individual’s age, sex, years of education, and APOE genotype, the method has better performance when predicting a PET-positive risk group for amyloid β deposition, and the higher the number of indicators identified, the better the predictive performance of the method may be.

[0106] Yet another embodiment of the present disclosure provides a composition for predicting a PET-positive risk group for amyloid β deposition, including a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual,

[0107] wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0108] The “individual,” “sample,” “single-nucleotide polymorphism,” “positron emission tomography (PET)-positive risk group for amyloid β deposition,” and the like may be within the above-described scope.

[0109] The composition for predicting a PET-positive risk group for amyloid β deposition was invented by using

European-based meta-GWAS results obtained from an inverse variance-weighted fixed-effect meta-analysis of various European GWAS results and analyzing predictive performance by including single-nucleotide polymorphisms in a P-value threshold range of the GWAS.

[0110] In an embodiment, the composition may predict a PET-positive risk group for amyloid β deposition with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0111] The preparation may be selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence including the single-nucleotide polymorphism or a protein encoded by the base sequence.

[0112] In an embodiment, the plurality of single-nucleotide polymorphisms may further include one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

[0113] When the plurality of single-nucleotide polymorphisms further include one or more single-nucleotide polymorphisms as described above, the composition has better performance when predicting a PET-positive risk group for amyloid β deposition, and the higher the number of single-nucleotide polymorphisms that are further included, the better the predictive performance of the composition may be.

[0114] In an embodiment, the composition may further include a preparation capable of identifying the APOE genotype of the individual in the sample.

[0115] In an embodiment, when the composition further includes a preparation capable of identifying the APOE genotype of the individual in the sample, the composition has better performance when predicting a PET-positive risk group for amyloid β deposition.

[0116] Yet another embodiment of the present disclosure provides a kit for predicting a PET-positive risk group for amyloid β deposition, including the composition.

[0117] The “positron emission tomography (PET)-positive risk group for amyloid β deposition” and “kit” may be within the above-described scope.

[0118] In an embodiment, the kit may predict a PET-positive risk group for amyloid β deposition with excellent accuracy by confirming the presence or absence of risk alleles of at least 11 single-nucleotide polymorphisms in a sample.

[0119] Yet another embodiment of the present disclosure provides a use of a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual for predicting a risk group for developing Alzheimer’s disease dementia or a risk group for early onset of Alzheimer’s symptoms,

[0120] wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0121] The “individual,” “sample,” “single-nucleotide polymorphism,” “Alzheimer’s disease dementia,” and the like may be within the above-described scope.

[0122] Yet another embodiment of the present disclosure provides a use of a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual for predicting a risk group for developing amnestic mild cognitive impairment,

[0123] wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0124] The “individual,” “sample,” “single-nucleotide polymorphism,” “amnestic mild cognitive impairment,” and the like may be within the above-described scope.

[0125] Yet another embodiment of the present disclosure provides a use of a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual for predicting a PET-positive risk group for amyloid β deposition,

[0126] wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431.

[0127] The “individual,” “sample,” “single-nucleotide polymorphism,” “positron emission tomography (PET)-positive risk group for amyloid β deposition,” and the like may be within the above-described scope.

Advantageous Effects

[0128] In an embodiment, the method makes it possible to accurately predict a risk group for developing Alzheimer’s disease dementia or a risk group for early onset of Alzheimer’s symptoms, or a risk group for developing amnestic mild cognitive impairment and/or a positron emission tomography (PET)-positive risk group for amyloid β deposition by using only at least 11 single-nucleotide polymorphisms, and the ability to predict the risk groups is further enhanced when up to and at most 39 additional single-nucleotide polymorphisms are used. In addition, the method further identifies age, sex, years of education, and APOE genotype as indicators, and thus the ability to predict the risk groups is further enhanced, and the risk groups can be predicted at an early stage with high accuracy.

DESCRIPTION OF DRAWINGS

[0129] FIGS. 1A and 1B are views illustrating the results of a comparison of principal components between Korean study populations and genome project populations.

[0130] FIG. 2 is a set of views illustrating the results of PRS distribution among study subjects according to genotype array.

[0131] FIG. 3 is a view illustrating an overview of the study dataset and analysis steps.

[0132] FIG. 4 is a set of views illustrating the results of distribution for Nagelkerke R^2 values of each population-based PRS across the single-nucleotide polymorphism selection threshold.

[0133] FIG. 5 is a view illustrating the scatter plots of beta coefficients for 39 single-nucleotide polymorphisms.

[0134] FIGS. 6A, 6B, and 6C are views illustrating the results of Forest plots of Alzheimer’s Disease Dementia (ADD)-related outcomes by PRS group and APOE ε4 status.

such as severe cerebral ischemia, cerebral infarction or brain tumors, or (iii) other medical or psychiatric diseases that may induce cognitive decline. All subjects were provided written informed consent, and the study was approved by the Institutional Review Board at each center.

TABLE 1

Characteristics	Validation dataset (n = 1,033)		Replication dataset (n = 379)		aMCI dataset
	CU (n = 479)	ADD (n = 554)	CU (n = 220)	ADD (n = 159)	aMCI (n = 222)
Age, mean (SD), year	70.7 ± 7.6	73.1 ± 10.0	67.8 ± 9.2	72.6 ± 8.6	73.0 ± 8.2
Female sex, no. (%)	282 (58.9)	348 (62.8)	139 (63.2%)	91 (57.2%)	109 (49.1)
Education, mean (SD), year	11.2 ± 4.9	10.4 ± 5.0	11.3 ± 4.6	9.7 ± 5.3	11.9 ± 4.7
APOE ε4 carrier, no. (%)	118 (24.6%)	314 (56.7%)	55 (25.0%)	74 (46.5%)	79 (35.6%)
Amyloid PET positivity, no. (%)	63 (14.0%)	479 (87.7%)	33 (15.0%)	119 (74.8%)	108 (49.3%)

Abbreviations: CU, cognitively unimpaired; aMCI, amnesic mild cognitive impairment; ADD, Alzheimer’s disease dementia; SD, standard deviation; and PET, positron emission tomography.

MODES OF THE INVENTION

2. Replication Dataset

[0135] Hereinafter, some embodiments of the present disclosure will be described in more detail through examples. However, these examples are provided only for exemplarily describing the present invention, and the scope of the present invention is not limited by these examples.

[0138] For a replication dataset, data from 379 Korean subjects was secured from 20 referral hospitals in Korea. Specifically, data on 125 subjects recruited from the biobank of the Chronic Cerebrovascular Disease Consortium from 2016 to 2018 was secured, and data on the remaining 254 subjects was secured from the PRECISION medicine platform for mild cognitive impairment based on Multi-omics, imaging, and Evidence-based R&BD (PREMIERM) cohort. Further, Alzheimer’s disease dementia (ADD) or cognitively unimpaired (CU) was included according to the same criteria of the validation dataset.

EXAMPLES

1. Validation and Amnesic Mild Cognitive Impairment (aMCI) Dataset

3. Genotype Analysis and Statistical Imputation

[0136] From January 2013 to July 2019, 1,255 Korean subjects were recruited from 14 hospitals in the Republic of Korea. Specifically, 954 participants were recruited from Samsung Seoul Hospital, 202 from the Korean Brain Aging Study for Early Diagnosis and Prediction of AD, and 99 from a multicenter clinical research platform study based on the Dementia Cohort (Table 1). Based on detailed neuropsychological test results, subjects diagnosed with Alzheimer’s disease dementia (ADD) or amnesic mild cognitive impairment (aMCI) or cognitively unimpaired (CU) were included, and the diagnosis of subjects at the most recent assessment point was used. Alzheimer’s disease dementia (ADD) was defined by the core clinical criteria for Alzheimer’s disease dementia (ADD) according to the National Institute on Aging-Alzheimer’s Association. Amnesic mild cognitive impairment (aMCI) was defined according to the following criteria modified from Peterson’s criteria: (i) normal activities of daily living performance, (ii) objective memory impairment, that is, performance ability below the 16th percentile of age-and education-matching norms in verbal or visual memory tests, and (iii) no dementia.

[0137] Subjects were excluded when they had (i) causative gene mutations for Alzheimer’s disease (AD) in known genes such as Presenilin-1 (PSEN1), Presenilin-2 (PSEN2) and amyloid-beta precursor protein (APP), (ii) structural abnormalities found by brain magnetic resonance imaging

[0139] DNA specimens were genotyped using Illumina Asian Screening Array BeadChips (ASA chips, CA, USA). A portion of the specimens (n=125) was genotyped using an Affymetrix custom-made Korea Biobank Array chip (KBA chip, Affymetrix, CA, USA), and quality control (QC) was performed on the data for two types of single-nucleotide polymorphisms (SNPs). SNPs were removed according to the following criteria: (i) call rate <98%, (ii) minor allele frequency (MAF) <1%, or (iii) genotype frequency that deviates significantly from Hardy-Weinberg equilibrium with a P-value of <10⁻⁶. After quality control (QC), the genotype data was directly subjected to genotype estimation for mutations for which no genotype was assigned, and statistical imputation methods were applied to combine the datasets of different genotype arrays (ASA chip and KBA chip). Statistical imputation of genotypes was performed using Minimac4 software with all reference haplotypes available in the Haplotype Reference Consortium (HRC-r1.1 2016) on the University of Michigan Imputation Server. Consequently, the present inventors performed quality control (QC) after statistical imputation with (i) a MAF <1% or (ii) low imputation quality (for imputed SNPs, R²<0.8). To identify appropriate combinations of the two genotype datasets, principal component analysis (PCA) was performed using EIGENSTRAT. In addition, PCA was performed on

1000 Genomes Project samples, and two genotype datasets were projected onto a PCA plot to confirm racial distinctions. Based on the genotype data, subjects were excluded according to the following criteria: (i) call rate <95%; (ii) sex mismatch; (iii) excess heterozygosity (± 5 standard deviations [SDs] from the mean); or (iv) one of the related pairs with second-degree consanguinity or less as estimated using KING software.

[0140] Comparison of principal components between the Korean study population and the 1,000 Genomes Project population revealed that there was racial overlap in principal component analysis (PCA) with the 1,000 Genomes Project dataset and data from other East Asian populations. However, there was no stratification by genotype array, and the distribution of polygenic risk scores (PRSs) among study subjects by genotype array did not differ significantly by genotype array (FIGS. 1A and 1B and FIG. 2).

4. Amyloid Positron Emission Tomography (PET)

[0141] A portion of subjects (n=1,214) from the validation and amnesic mild cognitive impairment (aMCI) datasets were subjected to amyloid positron emission tomography (PET), performed using a Discovery STE PET/computed tomography scanner (GE Medical Systems, Milwaukee, WI, USA). PET images were acquired for 20 minutes starting 90 minutes after intravenous injection of 18F-florbetaben or 18F-flutemetamol. amyloid β (A β) positivity or negativity was determined by well-trained nuclear medicine physicians using visual assessments of florbetaben PET or flutemetamol PET. Positivity for tracer uptake was assessed in four cortical regions (lateral temporal, frontal, parietal, and posterior cingulate cortices) for florbetaben PET and five cortical regions (lateral temporal, frontal, parietal, posterior cingulate cortices, and striatum) for flutemetamol PET. Amyloid PET positivity was defined as having at least one cortical region with evidence of positive uptake.

5. GWAS Summary Statistics

[0142] To investigate the transferability of polygenic risk scores (PRSs) in the Korean population, summary statistics generated in the European International Genomics of

Alzheimer's Project (IGAP) METAGWAS (11,480,632 SNPs in 21,982 AD patients and 41,944 controls) were utilized.

6. PRS Generation

[0143] Based on the data from previous studies, 3,877 SNPs surrounding APOE (chromosome 19, 44,400 to 46,500 kb, GRCH37/hg19) were excluded to derive a PRS independent of the APOE region, and PRSice-2 was used based on the previous European IGAP GWAS summary statistics to determine the best parameters (P-value threshold and linkage disequilibrium (LD) r^2 value) for PRS calculation. The P values and effect sizes of summary statistics were used to generate the best PRS model on the validation dataset (554 Koreans with Alzheimer's disease dementia (ADD) and 479 cognitively unimpaired (CU) controls). To derive the best model, testing was performed by including SNPs while varying a range of P-value thresholds (5×10^{-8} to 1.0) of the GWAS performed by Kunkle et al. Furthermore, the linkage disequilibrium (LD) r^2 (0.1-0.9) range within 1,000 kb was examined to investigate the critical value showing the largest Nagelkerke R^2 value calculated by logistic regression. Thereafter, the same SNPs and weighted values were used to replicate the PRS associations in an independent dataset of 379 specimens (159 Alzheimer's disease dementia (ADD) cases and 220 cognitively unimpaired (CU) controls) and an applied dataset of 222 patients with amnesic mild cognitive impairment (aMCI), and an overview showing the study datasets and analysis steps was shown (FIG. 3).

[0144] As a result, across various thresholds (P and LD values), the highest Nagelkerke R^2 value was identified at 0.020 ($P < 0.001$) for IGAP GWAS-based PRS, and the highest Nagelkerke R^2 value (0.020) was identified at P and LD values of 4.15×10^{-6} and 0.1, respectively, among various thresholds (FIG. 4). Further, 39 SNPs were selected from this threshold and the beta coefficients were used for PRS generation (Table 2), confirming a high correlation (Spearman correlation=0.533, $P < 0.001$) between the beta coefficients of the 39 SNPs calculated in the IGAP dataset and those in the validation dataset (FIG. 5).

TABLE 2

CHR	SNP	Nearest gene	Risk allele	EUR ¹ MAF	EAS ¹ MAF	IGAP		Korean (our validation dataset)	
						Beta ²	SE ²	Beta ³	SE ³
10	rs12358692	RP11-138I18.1	T	0.6572	0.3264	0.6429	0.0154	0.0133	0.0828
11	rs11218343	SORL1	C	0.0389	0.2977	-0.2053	0.0369	-0.0936	0.0826
2	rs6733839	BIN1	T	0.3921	0.4363	0.1693	0.0154	0.0048	0.0759
1	rs679515	CR1	T	0.1988	0.0284	0.1508	0.0183	0.1816	0.2186
1	rs1752684	CR1	A	0.2037	0.3526	0.1432	0.0178	0.0241	0.0789
20	rs6014724	CASS4	G	0.0905	0.3689	-0.1319	0.0259	-0.0474	0.0796
3	rs6805148	CLEC3B	C	0.0891	0.0812	-0.1293	0.0257	-0.0455	0.1387
8	rs1532276	CLU	T	0.3919	0.2031	-0.1266	0.0154	-0.0834	0.0901
11	rs1582763	MS4A6A	A	0.3593	0.133	-0.1232	0.0149	-0.0215	0.098
14	rs17125924	FERMT2	G	0.0954	0.2262	0.1222	0.0246	0.0298	0.0889
2	rs35832505	BIN1	C	0.1662	0.0562	0.1213	0.019	0.0318	0.1419
11	rs3851179	PICALM	T	0.3542	0.4103	-0.1198	0.0148	-0.0433	0.0783
2	rs35695568	RPL21P32	T	0.0963	0.1247	0.1152	0.0247	0.0645	0.113
11	rs56201148	MS4A6A	T	0.3823	0.2189	-0.1137	0.0146	-0.0955	0.0965
19	rs12151021	ABCA7	A	0.3205	0.5081	0.1071	0.0169	0.0267	0.0737
7	rs11767557	EPHA1-AS1	C	0.2011	0.1401	-0.1028	0.0182	-0.0759	0.1068

TABLE 2-continued

CHR	SNP	Nearest gene	Risk allele	EUR ¹ MAF	EAS ¹ MAF	IGAP		Korean (our validation dataset)	
						Beta ²	SE ²	Beta ³	SE ³
11	rs67472071	SPI1	A	0.3396	0.3026	-0.0981	0.0152	-0.0281	0.0773
19	rs3795065	ABCA7	C	0.3498	0.2548	0.0968	0.0171	-0.0152	0.0908
16	rs3752786	MTSS2	G	0.2213	0.1324	0.0964	0.0209	0.0354	0.1038
8	rs73223431	PTK2B	T	0.3352	0.2991	0.0936	0.0153	-0.0211	0.0906
14	rs11623019	SLC24A4	T	0.1934	0.4447	0.0913	0.0174	0.0615	0.0773
14	rs12590654	SLC24A4	A	0.3429	0.464	-0.0906	0.0157	-0.0668	0.0772
11	rs11039165	MADD	G	0.2668	0.0244	0.0894	0.0158	-0.0139	0.2328
16	rs28482811	GPRC5B	C	0.16	0.2671	-0.0872	0.0189	-0.0443	0.0894
6	rs3135348	BTNL2	A	0.4198	0.3705	0.0837	0.015	-0.0026	0.0818
6	rs9381563	AL355353.1	C	0.355	0.1979	0.0821	0.0148	0.0298	0.0977
6	rs9268112	TSBP1-AS1	A	0.3256	0.2629	0.0815	0.0159	-0.0208	0.0981
19	rs3865444	SIGLEC22P	A	0.3212	0.1733	-0.0804	0.0158	0.0035	0.0947
11	rs11230227	MS4A4E	A	0.3804	0.2584	0.0792	0.0153	0.0162	0.0843
6	rs9271375	HLA-DRB1	A	0.485	0.6226	-0.0789	0.0157	-0.0261	0.082
11	rs2293579	PSMC3	A	0.3868	0.3743	0.0771	0.0145	-0.0644	0.0825
8	rs7831810	GULOP	A	0.586	0.4013	-0.0765	0.0146	-0.0343	0.0793
5	rs11168036	PFDN1	T	0.4918	0.4339	0.0754	0.0143	0.0734	0.0768
11	rs598561	SLC25A1P1	A	0.4873	0.0963	0.0747	0.0143	0.0508	0.1088
21	rs3017432	ADAMTS1	T	0.3967	0.7336	-0.0735	0.0151	-0.0481	0.0825
17	rs2526378	BZRAP1	A	0.4551	0.5304	0.0717	0.0145	-0.0026	0.0764
19	rs8111708	ELL	G	0.3562	0.2523	0.0696	0.0151	0.0667	0.0869
7	rs7805776	EPHA1-AS1	A	0.5248	0.3732	-0.0695	0.0148	-0.0237	0.0776
6	rs12197146	CD2AP	C	0.4873	0.1731	-0.0674	0.0144	-0.0164	0.0933

[0145] 1MAF for non-Finnish EUR and EAS samples from the Genome Count Database. (gnomAD version 2.1.1, <https://gnomad.broadinstitute.org>)

[0146] Statistical values were obtained from ²a previous studies and ³our validation dataset.

[0147] Abbreviations: IGAP, International Genomics of Alzheimer's Project; CHR, chromosomal; SNP, single-nucleotide polymorphism; SE, standard error; MAF, minor allele frequency; PRS, polygenic risk score; EUR, European; EAS, East Asian.

7. Validation and Replication of PRS for Alzheimer's Disease Dementia (ADD) Diagnosis

[0148] After the PRS for each subject was calculated, a logistic regression analysis was performed to determine whether the PRS derived from summary statistics for Alzheimer's disease (AD) risk based on the European population was associated with Alzheimer's disease dementia (ADD) diagnosis in the validation examination and replication datasets after correcting age, sex, years of education, APOE ε4 carrier status, and the first four principal components (PCs) of genetic ancestry using a multivariate logistic regression model. In addition, to confirm that the association between Alzheimer's disease dementia (ADD) diagnosis

and the PRS differed by APOE ε4 carrier status, the same analysis was performed after stratifying subjects into APOE ε4 carrier and non-carrier groups, a PRS was developed based on previous European IGAP GWAS results, and the PRS predictive performance was compared. Odds ratio (OR) and P values were calculated using a multivariate logistic regression analysis (OR per standard deviation increase in standardized PRS).

[0149] As a result, it was confirmed that high PRS was associated with an increase in the risk of Alzheimer's disease dementia (ADD) after correcting the effects of age, sex, education, and APOE ε4 status, and that PRS was associated with the risk of Alzheimer's disease dementia (ADD) in both APOE ε4 carrier (odds ratio [OR]=2.73, 95% CI=1.53 to 4.97, P=0.001) and non-carrier (OR=1.70, 95% CI=1.14-2.59, P=0.01) groups. Furthermore, it was confirmed that higher PRS was significantly associated with the increased risk of amnesic mild cognitive impairment (aMCI) and amyloid β (Aβ) deposition in the brain (Table 3).

TABLE 3

ADD diagnosis ¹				aMCI diagnosis ²		Aβ PET deposition ³	
Dataset							
Validation			Replication		Application		Application
Diagnosis, no.							
CU (n = 479) vs. ADD (n = 554)			CU (n = 220) vs. ADD (n = 159)		CU (n = 479) vs. aMCI (n = 220)		Aβ (-) (n = 564) vs. Aβ (+) (n = 650)
OR (95% CI)		P	OR (95% CI)		OR (95% CI)		OR (95% CI)
PRS	1.95 (1.40-2.72)	<0.001	1.85 (1.05-3.32)	0.036	1.74 (1.16-2.64)	0.008	1.81 (1.32-2.48)
							<0.001

8. Application of PRS in Various Phenotypes

[0150] A multivariate logistic regression analysis was performed on subjects with amnesic mild cognitive impairment (aMCI) to assess whether the PRS predicted aMCI independently of age, sex, years of education, APOE ϵ 4 carrier status, and the first four principal components (PCs) of genetic ancestry. In a portion of subjects (n=1,214) who were also subjected to amyloid (A β) PET, a logistic regression analysis was also performed to assess whether the PRS predicted amyloid (A β) positivity, and the effects of age, sex, years of education, and APOE ϵ 4 carrier status, which had been subjected to amyloid β (A β) PET, were adjusted.

[0151] Further, to test the clinical utility of the PRS, a multivariate logistic model to predict Alzheimer's disease dementia (ADD) diagnosis for each subject was developed, and an area under curve (AUC) was measured to assess the performance of the logistic model. For internal validation, 10-fold cross-validation was performed with 100 repetitions using validation data. The mean AUC with the 95% confidence interval (CI) of the models was reported, and the AUCs of models were compared by applying the DeLong test.

[0152] As a result, in the prediction model, the model including only clinical factors (age, sex, and years of education) (Model 1) showed an AUC of 0.581 (95% CI=0.578 to 0.585), and it was confirmed that the predictive performance increased after integrating APOE ϵ 4 status into Model 1 (Model 2) (AUC=0.693; 95% CI=0.690 to 0.696). In addition, it was confirmed that when the model included PRS (Model 3), the predictive performance was remarkably improved compared to Model 2 (AUC=0.706; 95% CI=0.703-0.709, $P=6.30 \times 10^{-9}$) (FIGS. 6A to 6C).

[0153] Furthermore, subjects were stratified based on quartiles of PRS, it was evaluated whether PRS could also be used for risk stratification in addition to APOE ϵ 4 genotype, and it was evaluated whether subjects with a higher PRS exhibited earlier development of Alzheimer's disease (AD) than those with a lower PRS. Further, Cox regression analysis was performed by employing age at last clinical visit or age at onset of Alzheimer's disease (AD) as a time variable and Alzheimer's disease (AD) as a status variable.

[0154] As a result, it was confirmed that when PRS and APOE ϵ 4 status were combined, in both the APOE ϵ 4 carrier and non-carrier, the risks of Alzheimer's disease dementia (ADD), amyloid β (A β) deposition, and early onset of Alzheimer's disease dementia increased in a stepwise man-

ner according to PRS quartile (FIGS. 6A to 6C). In particular, compared to the APOE ϵ 4 non-carriers in the low PRS group, the APOE ϵ 4 carriers in the very high PRS group were confirmed to have a 6.73-fold (95% CI=3.99 to 11.75), 15.04-fold (95% CI=8.45 to 28.13), and 2.74-fold (95% CI=1.88 to 4.00) higher risk of Alzheimer's disease dementia (ADD), amyloid β (A β) deposition, and age at initial onset of symptoms, respectively. This coincides with previous findings showing that PRS is associated with Alzheimer's disease pathology (A β deposition, tau and neurodegeneration), and it was confirmed that it is important for predicting prognosis and selecting patients for clinical trials of anti-A β therapies to identify patients with amyloid β (A β) deposition. Currently, diagnostic tools for measuring amyloid β (A β) deposition are either invasive (cerebrospinal fluid examination) or expensive (PET). The study results highlight that genetic data (PRS and APOE ϵ 4 status) obtained from less invasive methods (blood or saliva specimen assessment) can be used to pre-screen for amyloid β (A β) positivity.

[0155] In addition, it was confirmed that patients with a high PRS were more likely to develop Alzheimer's disease dementia (ADD) symptoms at a young age. The mean age at onset of symptoms was about 3.7 years younger in the very high PRS group than in the low PRS group. It is well known that APOE ϵ 4 is associated with the onset of early symptoms of Alzheimer's disease dementia (ADD), and through the results of the present inventors, it was confirmed that PRS further accelerates the age of onset of symptoms beyond the effect of APOE ϵ 4.

9. Statistical Analysis

[0156] Categorical and continuous variables for demographic and clinical characteristics of subjects according to PRS quantiles are presented as counts (%) and means (SDs), respectively. P values were obtained using a chi-square test in analysis of variance for categorical and continuous variables (Table 4). Two-sided P values were reported, and a P value <0.05 was defined as statistically significant. Furthermore, all statistical analyses and results were visualized using PLINK 1.90, R version 3.6.1 (R Project for Statistical Computing) and MATLAB.

TABLE 4

	Low PRS group (n = 314)	Intermediate PRS group (n = 314)	High PRS group (n = 314)	Very high PRS group (n = 313)	P
Age, mean (SD), year	72.6 ± 8.9	72.6 ± 8.8	71.9 ± 9.1	71.7 ± 8.7	0.146
Education, mean (SD), year	11.2 ± 4.9	10.9 ± 5.0	10.7 ± 5.1	11.1 ± 4.9	0.761
Female sex, no. (%)	172 (54.8)	182 (58.0)	197 (62.7)	188 (60.1)	0.221
APOE ε4 carrier, no. (%)	117 (37.3)	129 (41.1)	132 (42.0)	133 (42.5)	0.531
Amyloid positivity, no. (%)	139 (45.7)	161 (53.1)	169 (55.6)	181 (59.7)	0.005
Age at ADD symptom onset, mean (SD), year	69.0 ± 9.9	68.1 ± 9.8	66.5 ± 10.4	65.3 ± 9.7	0.012
Diagnosis, no. (%)					0.005
CU	150 (47.8%)	115 (36.6%)	112 (35.7%)	102 (32.6%)	
aMCI	51 (16.2%)	59 (18.8%)	53 (16.9%)	59 (18.8%)	
ADD	113 (36.0%)	140 (44.6%)	149 (47.5%)	152 (48.6%)	

Abbreviations: CU, cognitively unimpaired; aMCI, amnesic mild cognitive impairment; ADD, Alzheimer's disease dementia; PRS, polygenic risk score; SD, standard deviation.

10. Prediction Model for Risk Group for Alzheimer's Disease Dementia (ADD)

area under curve (AUC): 0.5678; Nagelkerke R²: 0.0203; and P-value<0.0001 were obtained.

(1) Prediction Model for Risk Group for Alzheimer's Disease Dementia (ADD) in Consideration of Coefficients for Each SNP

[0157] In consideration of the actual coefficients for each SNP (Table 5), a calculation formula corresponding to the following Mathematical Formula 1 that can be used for a prediction model for the risk group was obtained.

[Mathematical Formula 1]

(Model 1) PRS (39 SNPs) =

$$\begin{aligned}
 &rs12358692*0.6429 + rs11218343* -0.2053 + rs6733839*0.1693 + \\
 &rs679515*0.1508 + rs1752684*0.1432 + rs6014724* -0.1319 + \\
 &rs6805148* -0.1293 + rs1532276* -0.1266 + rs1582763* - \\
 &0.1232 + rs17125924*0.1222 + rs35832505*0.1213 + \\
 &rs3851179* -0.1198 + rs35695568*0.1152 + rs56201148* - \\
 &0.1137 + rs12151021*0.1071 + rs11767557* -0.1028 + \\
 &rs67472071* -0.0981 + rs3795065*0.0968 + rs3752786*0.0964 + \\
 &rs73223431*0.0936 + rs11623019*0.0913 + rs12590654* -0.0906 + \\
 &rs11039165*0.0894 + rs28482811* -0.0872 + rs3135348*0.0837 + \\
 &rs9381563*0.0821 + rs9268112*0.0815 + rs3865444* -0.0804 + \\
 &rs11230227*0.0792 + rs9271375* -0.0789 + rs2293579*0.0771 + \\
 &rs7831810* -0.0765 + rs11168036*0.0754 + rs598561*0.0747 + \\
 &rs3017432* -0.0735 + rs2526378*0.0717 + rs8111708*0.0696 + \\
 &rs7805776* -0.0695 + rs12197146* -0.0674
 \end{aligned}$$

[0158] Further, <Model 1> the predictive performance of the PRS model alone for the risk group for Alzheimer's disease dementia (ADD) was analyzed, and the values of the

TABLE 5

	Constituent element	Coefficient
1	rs12358692	0.6429
2	rs11218343	-0.2053
3	rs6733839	0.1693
4	rs679515	0.1508
5	rs1752684	0.1432
6	rs6014724	-0.1319
7	rs6805148	-0.1293
8	rs1532276	-0.1266
9	rs1582763	-0.1232
10	rs17125924	0.1222
11	rs35832505	0.1213
12	rs3851179	-0.1198
13	rs35695568	0.1152
14	rs56201148	-0.1137
15	rs12151021	0.1071
16	rs11767557	-0.1028
17	rs67472071	-0.0981
18	rs3795065	0.0968
19	rs3752786	0.0964
20	rs73223431	0.0936
21	rs11623019	0.0913
22	rs12590654	-0.0906
23	rs11039165	0.0894
24	rs28482811	-0.0872
25	rs3135348	0.0837
26	rs9381563	0.0821
27	rs9268112	0.0815
28	rs3865444	-0.0804
29	rs11230227	0.0792
30	rs9271375	-0.0789
31	rs2293579	0.0771
32	rs7831810	-0.0765
33	rs11168036	0.0754
34	rs598561	0.0747
35	rs3017432	-0.0735
36	rs2526378	0.0717
37	rs8111708	0.0696
38	rs7805776	-0.0695
39	rs12197146	-0.0674

(2) Prediction Model for Risk Group for Alzheimer's Disease Dementia (ADD) when Further Including Four Factors (Age, Sex, Years of Education, and APOE ε4)

[0159] When each of the four factors (age, sex, years of education, and APOE ε4) was included, a calculation formula used in the prediction model for the risk group for Alzheimer's disease dementia (ADD) was obtained. Definitions for each factor; PRS is a genetic risk score for each individual calculated by the model suggested above; age is assigned in years; sex is defined as 1 for males and 2 for females; years of education are assigned in years; and for the APOE genotype among the indicators of the individual, it is possible to further include obtaining a score for each indicator by assigning a score of 0 for ε2/ε2, ε2/ε3 and ε3/ε3 and a score of 1 for ε2/ε4, ε3/ε4 and ε3/ε4. A calculation formula corresponding to the following Mathematical Formula 2, which can be used for a regression model with PRS scores and four additional factors as dependent terms, was obtained.

[Mathematical Formula 2]

$$PRS \text{ (including 4 factors)} = (\text{Model 1}) PRS * 0.66119 + \text{age} * 0.02879 + \text{sex} * 0.03618 + \text{education} * -0.02615 + APOE\epsilon 4 * 1.37120$$

[0160] In addition, the predictive performance of the PRS <Model 1>, in which the four factors were considered together, for the risk group for Alzheimer's disease dementia (ADD) was analyzed, and the values of the area under curve (AUC): 0.7101; Nagelkerke R²: 0.1756; and P-value=0.0002 were obtained.

(3) Predictive Ability of 31 SNPs

[0161] Model 2 is an additional selection process from 39 SNPs used in the construction of Model 1, and a PRS model was constructed by selecting 31 SNPs with the same association β coefficient direction between the Alzheimer's disease dementia (ADD) patient group and the control group from Korean data and European data (FIG. 5). The difference between the models is the difference due to the presence or absence of factors, and the estimated coefficients for each factor are the same (Table 6). Furthermore, in consideration of the coefficients for the selected 31 SNPs, a calculation formula corresponding to the following Mathematical Formula 3 that can be used for a prediction model for the risk group was obtained.

[Mathematical Formula 3]

$$\begin{aligned} (\text{Model 2}) PRS \text{ (31 SNPs with feature selection)} = & rs12358692 * 0.6429 + rs11218343 * -0.2053 + rs6733839 * 0.1693 + \\ & rs679515 * 0.1508 + rs1752684 * 0.1432 + rs6014724 * -0.1319 + \\ & rs6805148 * -0.1293 + rs1532276 * -0.1266 + rs1582763 * - \\ & 0.1232 + rs17125924 * 0.1222 + rs35832505 * 0.1213 + rs3851179 * - \\ & 0.1198 + rs35695568 * 0.1152 + rs56201148 * -0.1137 + \\ & rs12151021 * 0.1071 + rs11767557 * -0.1028 + rs67472071 * - \end{aligned}$$

-continued

$$\begin{aligned} & 0.0981 + rs3752786 * 0.0964 + rs11623019 * 0.0913 + rs12590654 * - \\ & 0.0906 + rs28482811 * -0.0872 + rs9381563 * 0.0821 + \\ & rs11230227 * 0.0792 + rs9271375 * -0.0789 + rs7831810 * -0.0765 + \\ & rs11168036 * 0.0754 + rs598561 * 0.0747 + rs3017432 * -0.0735 + \\ & rs8111708 * 0.0696 + rs7805776 * -0.0695 + rs12197146 * -0.0674 \end{aligned}$$

[0162] As a result of analyzing the predictive performance of the PRS model for the risk group for Alzheimer's disease dementia (ADD), it was confirmed that the model of additionally selected 31 SNPs had a more excellent ability to predict the risk group for Alzheimer's disease dementia (ADD) than the model of 39 SNPs, and the values of the area under curve (AUC): 0.5787; Nagelkerke R²: 0.02771; and P-value<0.0001 were obtained.

[0163] As a result of analyzing the predictive performance of the PRS <Model 1>, in which the four factors were considered together, for the risk group for Alzheimer's disease dementia (ADD), the values of the area under curve (AUC): 0.7143; Nagelkerke R²: 0.1804; and P-value<0.0001 were obtained.

TABLE 6

	Constituent element	Coefficient
1	rs12358692	0.6429
2	rs11218343	-0.2053
3	rs6733839	0.1693
4	rs679515	0.1508
5	rs1752684	0.1432
6	rs6014724	-0.1319
7	rs6805148	-0.1293
8	rs1532276	-0.1266
9	rs1582763	-0.1232
10	rs17125924	0.1222
11	rs35832505	0.1213
12	rs3851179	-0.1198
13	rs35695568	0.1152
14	rs56201148	-0.1137
15	rs12151021	0.1071
16	rs11767557	-0.1028
17	rs67472071	-0.0981
18	rs3752786	0.0964
19	rs11623019	0.0913
20	rs12590654	-0.0906
21	rs28482811	-0.0872
22	rs9381563	0.0821
23	rs11230227	0.0792
24	rs9271375	-0.0789
25	rs7831810	-0.0765
26	rs11168036	0.0754
27	rs598561	0.0747
28	rs3017432	-0.0735
29	rs8111708	0.0696
30	rs7805776	-0.0695
31	rs12197146	-0.0674

(4) Performance of 28 PRS Models

[0164] The performance of the PRS models constructed by sequentially excluding 39 SNPs (while reducing ones in a less significant sequence according to the significance level of P-value) and 28 types of PRS models were obtained.

[0165] In the experiment, several PRS models were constructed while sequentially removing SNPs with relatively high significance levels (P-values) from the PRS (39 SNPs) model, each model was evaluated alone or as a model in which four well-known factors (age, sex, years of education, and APOE ϵ 4) were considered, and performance levels were compared with the AUC and significance level P-value.

[0166] As a result of the experiment, it was confirmed that a total of 28 detailed PRS models, from the PRS (38 SNPs) model in which one SNP was excluded from the <Model 1> PRS to the PRS using 11 SNPs, acted as factors which predict a risk group for Alzheimer's disease dementia (ADD) while securing statistical significance (Table 7).

TABLE 7

No.	SNP	Nearest Gene	Risk allele	Beta (IGAP2019)	Association P-value (IGAP2019)	AUC performance and PRS significance level of PRS single model	Final model AUC performance and significance level of PRS after correction of five factors
1	rs6733839	BIN1	T	0.1693	4.02E-28	AUC = 0.5043, P = 0.8136	AUC = 0.6978, P = 0.8888
2	rs1582763	MS4A6A	A	-0.1232	1.19E-16	AUC = 0.5089, P = 0.4870	AUC = 0.6982, P = 0.6347
3	rs679515	CR1	T	0.1508	1.56E-16	AUC = 0.5191, P = 0.2041	AUC = 0.6988, P = 0.3380
4	rs1532276	CLU	T	-0.1266	1.73E-16	AUC = 0.5285, P = 0.0738	AUC = 0.7001, P = 0.1627
5	rs3851179	PICALM	T	-0.1198	5.81E-16	AUC = 0.5279, P = 0.0853	AUC = 0.7012, P = 0.1847
6	rs1752684	CR1	A	0.1432	9.39E-16	AUC = 0.5363, P = 0.0304	AUC = 0.7031, P = 0.0828
7	rs56201148	MS4A6A	T	-0.1137	6.00E-15	AUC = 0.5370, P = 0.0276	AUC = 0.7033, P = 0.0586
8	rs67472071	SPI1	A	-0.0981	1.14E-10	AUC = 0.5392, P = 0.0293	AUC = 0.7033, P = 0.0639
9	rs35832505	BIN1	T	-0.1213	1.82E-10	AUC = 0.5393, P = 0.0260	AUC = 0.7036, P = 0.0501
10	rs12151021	ABCA7	A	0.1071	2.56E-10	AUC = 0.5386, P = 0.0339	AUC = 0.7037, P = 0.0547
11	rs73223431	PTK2B	T	0.0936	8.34E-10	AUC = 0.5463, P = 0.0135	AUC = 0.7046, P = 0.0298
12	rs12590654	SLC24A4	A	-0.0906	8.73E-09	AUC = 0.5465, P = 0.0132	AUC = 0.7045, P = 0.0307
13	rs11039165	MADD	A	-0.0894	1.50E-08	AUC = 0.5504, P = 0.0068	AUC = 0.7057, P = 0.0152
14	rs11767557	EPHA1-AS1	T	0.1028	1.56E-08	AUC = 0.5452, P = 0.0126	AUC = 0.7043, P = 0.0265
15	rs3795065	ABCA7	T	-0.0968	1.63E-08	AUC = 0.5442, P = 0.0140	AUC = 0.7045, P = 0.0269
16	rs3135348	BTNL2	A	0.0837	2.34E-08	AUC = 0.5485, P = 0.0025	AUC = 0.7060, P = 0.0074
17	rs11218343	SORL1	T	0.2053	2.63E-08	AUC = 0.5495, P = 0.0025	AUC = 0.7058, P = 0.0073
18	rs9381563	AL355353.1	T	-0.0821	2.93E-08	AUC = 0.5454, P = 0.0056	AUC = 0.7046, P = 0.0112
19	rs12358692	RP11-138118.1	T	0.6429	4.65E-08	AUC = 0.5528, P = 0.0022	AUC = 0.7063, P = 0.0044
20	rs2293579	PSMC3	A	0.0771	1.14E-07	AUC = 0.5537, P = 0.0021	AUC = 0.7060, P = 0.0053
21	rs11168036	PFDN1	T	0.0754	1.43E-07	AUC = 0.5591, P = 0.0008	AUC = 0.7077, P = 0.0022
22	rs7831810	GULOP	A	-0.0765	1.58E-07	AUC = 0.5594, P = 0.0006	AUC = 0.7078, P = 0.0020
23	rs11623019	SLC24A4	T	0.0913	1.66E-07	AUC = 0.5570, P = 0.0008	AUC = 0.7073, P = 0.0023
24	rs598561	SLC25A1P1	A	0.0747	1.87E-07	AUC = 0.5561, P = 0.0011	AUC = 0.7065, P = 0.0032
25	rs11230227	MS4A4E	A	0.0792	2.27E-07	AUC = 0.563, P = 0.0004	AUC = 0.7077, P = 0.0013
26	rs9268112	TSBP1-AS1	A	0.0815	2.94E-07	AUC = 0.5624, P = 0.0005	AUC = 0.7078, P = 0.0016
27	rs6014724	CASS4	A	0.1319	3.65E-07	AUC = 0.5624, P = 0.0004	AUC = 0.7077, P = 0.0013
28	rs3865444	SIGLEC22P	A	-0.0804	3.93E-07	AUC = 0.5619, P = 0.0003	AUC = 0.7074, P = 0.0011

TABLE 7-continued

No.	SNP	Nearest Gene	Risk allele	Beta (IGAP2019)	Association P-value (IGAP2019)	AUC performance and PRS significance level of PRS single model	Final model AUC performance and significance level of PRS after correction of five factors
29	rs9271375	HLA-DRB1	A	-0.0789	4.73E-07	AUC = 0.5628, P = 0.0003	AUC = 0.7083, P = 0.0008
30	rs6805148	CLEC3B	A	0.1293	4.77E-07	AUC = 0.5608, P = 0.0003	AUC = 0.7076, P = 0.0009
31	rs17125924	FERMT2	A	-0.1222	6.62E-07	AUC = 0.5636, P = 0.0002	AUC = 0.7079, P = 0.0005
32	rs2526378	BZRAP1	A	0.0717	7.74E-07	AUC = 0.5634, P = 0.0002	AUC = 0.7084, P = 0.0004
33	rs3017432	ADAMTS1	T	-0.0735	1.05E-06	AUC = 0.5642, P = 0.0002	AUC = 0.7089, P = 0.0003
34	rs7805776	EPHA1-AS1	A	-0.0695	2.61E-06	AUC = 0.5658, P < 0.0001	AUC = 0.7089, P = 0.0003
35	rs12197146	CD2AP	T	0.0674	2.85E-06	AUC = 0.5646, P < 0.0001	AUC = 0.7088, P = 0.0003
36	rs35695568	RPL21P32	T	0.1152	3.20E-06	AUC = 0.5672, P < 0.0001	AUC = 0.7096, P = 0.0002
37	rs28482811	GPRC5B	T	0.0872	3.94E-06	AUC = 0.5696, P < 0.0001	AUC = 0.7101, P = 0.0001
38	rs8111708	ELL	A	-0.0696	3.95E-06	AUC = 0.5680, P < 0.0001	AUC = 0.7099, P = 0.0002
Following <Model 1> PRS (39 SNPs)							
39	rs3752786	MTSS2	A	-0.0964	4.07E-06	AUC = 0.5678, P < 0.0001	AUC = 0.7101, P = 0.0002

[0167] A calculation formula corresponding to the following Mathematical Formula 4 that can be used for a prediction model for the risk group was obtained.

[Mathematical Formula 4]

(Model 3) PRS (11 SNPs) =

$$\begin{aligned} &rs6733839*0.1693 + rs1582763* -0.1232 + rs679515*0.1508 + \\ &rs1532276* -0.1266 + rs3851179* -0.1198 + rs1752684*0.1432 + \\ &rs56201148* -0.1137 + rs67472071* -0.0981 + rs35832505* - \\ &0.1213 + rs12151021*0.1071 + rs73223431*0.0936 \end{aligned}$$

[0168] It is possible to construct up to the following <Model 30> PRS while adding genetic factors one by one to the above mathematical formula. For example, Models 4 to 30 may be constructed as follows.

<Model 4> PRS (11 SNPs)

[0169] It is possible to construct models sequentially from

$$\begin{aligned} &=rs6733839*0.1693+rs1582763*-0.1232+rs679515*0.1508+rs1532276*-0.1266+rs3851179*-0.1198+ \\ &rs1752684*0.1432+rs56201148*-0.1137+rs67472071*-0.0981+rs35832505*-0.1213+rs12151021*0.1071+ \\ &rs73223431*0.0936+rs12590654*-0.0906, \end{aligned}$$

<Model 30> PRS (38 SNPs)

$$\begin{aligned} &\text{to} = rs6733839*0.1693 + rs1582763* -0.1232 + rs679515*0.1508 + \\ &rs1532276* -0.1266 + rs3851179* -0.1198 + rs1752684*0.1432 + \\ &rs56201148* -0.1137 + rs67472071* -0.0981 + rs35832505* - \\ &0.1213 + rs12151021*0.1071 + rs73223431*0.0936 + rs12590654* - \\ &0.0906 + rs11039165* -0.0894 + rs11767557*0.1028 + \\ &rs3795065* -0.0968 + rs3135348*0.0837 + rs11218343*0.2053 + \\ &rs9381563* -0.0821 + rs12358692*0.6429 + rs2293579*0.0771 + \\ &rs11168036*0.0754 + rs7831810* -0.0765 + rs11623019*0.0913 + \\ &rs598561*0.0747 + rs11230227*0.0792 + rs9268112*0.0815 + \\ &rs6014724*0.1319 + rs3865444* -0.0804 + rs9271375* -0.0789 + \\ &rs6805148*0.1293 + rs17125924* -0.1222 + rs2526378*0.0717 + \\ &rs3017432* -0.0735 + rs7805776* -0.0695 + rs12197146*0.0674 + \\ &rs35695568*0.1152 + rs28482811*0.0872 + rs8111708* -0.0696. \end{aligned}$$

1. A method for providing information for predicting a risk group, the method comprising: bringing a sample isolated from an individual in contact with a preparation capable of identifying the presence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs); and determining the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms in the sample, wherein the plurality of single-nucleotide polymorphisms comprise rs6733839, rs1582763, rs679515, rs1532276,

rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431, and

wherein the risk group is selected from the group consisting of a risk group for developing Alzheimer's disease dementia, a risk group for early onset of Alzheimer's symptoms, a risk group for developing amnesic mild cognitive impairment, and a positron emission tomography (PET)-positive risk group for amyloid β deposition.

2. The method of claim 1, wherein the plurality of single-nucleotide polymorphisms further comprise one or more single-nucleotide polymorphisms selected from the group consisting of rs12358692, rs11218343, rs6014724, rs6805148, rs17125924, rs35695568, rs11767557, rs3795065, rs3752786, rs11623019, rs12590654, rs11039165, rs28482811, rs3135348, rs9381563, rs9268112, rs3865444, rs11230227, rs9271375, rs2293579, rs7831810, rs11168036, rs598561, rs3017432, rs2526378, rs8111708, rs7805776, and rs12197146.

3. The method of claim 2, further comprising obtaining a score for a single-nucleotide polymorphism by assigning a score of 1 to a single-nucleotide polymorphism determined to indicate the presence of a risk allele in the sample among the plurality of single-nucleotide polymorphisms, wherein among the plurality of single-nucleotide polymorphisms, a single-nucleotide polymorphism determined to be absent in the sample is assigned a score of 0.

4. The method of claim 3, further comprising obtaining a first polygenic risk score (PRS) value by multiplying the assigned score for the single-nucleotide polymorphism by a coefficient (β) assigned for each of the following single-nucleotide polymorphisms, and adding all the multiplied values,

wherein the coefficient of rs6733839 is 0.1693, the coefficient of rs1582763 is -0.1232, the coefficient of rs679515 is 0.1508, the coefficient of rs1532276 is -0.1266, the coefficient of rs3851179 is -0.1198, the coefficient of rs1752684 is 0.1432, the coefficient of rs56201148 is -0.1137, the coefficient of rs67472071 is -0.0981, the coefficient of rs35832505 is 0.1213, the coefficient of rs12151021 is 0.1071, the coefficient of rs73223431 is 0.0936, the coefficient of rs12358692 is 0.6429, the coefficient of rs11218343 is -0.2053, the coefficient of rs6014724 is -0.1319, the coefficient of rs6805148 is -0.1293, the coefficient of rs17125924 is 0.1222, the coefficient of rs35695568 is 0.1152, the coefficient of rs11767557 is -0.1028, the coefficient of rs3795065 is 0.0968, the coefficient of rs3752786 is 0.0964, the coefficient of rs11623019 is 0.0913, the coefficient of rs12590654 is -0.0906, the coefficient of rs11039165 is 0.0894, the coefficient of rs28482811 is -0.0872, the coefficient of rs3135348 is 0.0837, the coefficient of rs9381563 is 0.0821, the coefficient of rs9268112 is 0.0815, the coefficient of rs3865444 is -0.0804, the coefficient of rs11230227 is 0.0792, the coefficient of rs9271375 is -0.0789, the coefficient of rs2293579 is 0.0771, the coefficient of rs7831810 is -0.0765, the coefficient of rs11168036 is 0.0754, the coefficient of rs598561 is 0.0747, the coefficient of rs3017432 is -0.0735, the coefficient of rs2526378 is 0.0717, the coefficient of rs8111708 is 0.0696, the coefficient of rs7805776 is -0.0695, and the coefficient of rs12197146 is -0.0674.

5. The method of claim 4, further comprising determining that, when the first PRS value is higher than the first PRS value of an individual not classified in a given risk group, the individual is in the given risk group, wherein the given risk group is selected from the group consisting of a high risk group for developing Alzheimer's disease dementia, a high risk group for early onset of Alzheimer's symptoms, a high risk group for developing amnesic mild cognitive impairment, and a PET-positive high risk group for amyloid β deposition.

6. The method of claim 5, further comprising identifying one or more indicators selected from the group consisting of the individual's age, sex, years of education, and APOE genotype.

7. The method of claim 6, further comprising obtaining a score for each indicator by assigning a score based on a number of years in the case of the age and years of education among the indicators of the individual,

assigning a score of 1 for males and a score of 2 for females in the case of sex among the indicators of the individual, and

assigning a score of 0 for $\epsilon 2/\epsilon 2$, $\epsilon 2/\epsilon 3$, and $\epsilon 3/\epsilon 3$ and a score of 1 for $\epsilon 2/\epsilon 4$, $\epsilon 3/\epsilon 4$, and $\epsilon 4/\epsilon 4$ in the case of APOE genotype among the indicators of the individual.

8. The method of claim 7, further comprising obtaining a second PRS value by multiplying the assigned score for each indicator by a coefficient (β) assigned for each of the following indicators, and adding the first PRS value and a coefficient (β) assigned for the following first PRS value to the multiplied values,

wherein the coefficient of the age is 0.02879, the coefficient of the sex is 0.03618, the coefficient of the year of education is -0.02615, the coefficient of the APOE genotype is 1.3712, and the coefficient of the first PRS value is 0.66119.

9. The method of claim 8, further comprising determining that, when the second PRS value is higher than the second PRS value of an individual not classified in a given risk group, the individual is in the given risk group, wherein the given risk group is selected from the group consisting of a high risk group for developing Alzheimer's disease dementia, a high risk group for early onset of Alzheimer's symptoms, a high risk group for developing amnesic mild cognitive impairment, and a PET-positive high risk group for amyloid β deposition.

10. (canceled)

11. The method of claim 1, wherein the preparation is selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence comprising the single-nucleotide polymorphism or a protein encoded by the base sequence.

12. A composition for predicting a risk group, comprising a preparation capable of confirming the presence or absence of risk alleles of a plurality of single-nucleotide polymorphisms (SNPs) in a sample isolated from an individual,

wherein the plurality of single-nucleotide polymorphisms are rs6733839, rs1582763, rs679515, rs1532276, rs3851179, rs1752684, rs56201148, rs67472071, rs35832505, rs12151021, and rs73223431, and

wherein the risk group is selected from the group consisting of a risk group for developing Alzheimer's disease dementia, a risk group for early onset of Alzheimer's symptoms, a risk group for developing

amnesic mild cognitive impairment, and a positron emission tomography (PET)-positive risk group for amyloid β deposition.

13. The composition of claim **12**, wherein the preparation is selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence comprising the single-nucleotide polymorphism or a protein.

14. A kit for predicting a risk group, comprising the composition of claim **12**,

wherein the risk group is selected from the group consisting of a risk group for developing Alzheimer's disease dementia, a risk group for early onset of Alzheimer's symptoms, a risk group for developing amnesic mild cognitive impairment, and a positron emission tomography (PET)-positive risk group for amyloid β deposition.

15-23. (canceled)

24. The kit of claim **14**, wherein the preparation is selected from the group consisting of a primer, a probe, an aptamer, an antibody, a peptide, and combinations thereof, capable of specifically binding to a base sequence comprising the single-nucleotide polymorphism or a protein.

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